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A Signal Detection–Based Confidence–Similarity Model of Face Matching

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Face matching consists of the ability to decide whether two face images (or more) belong to the same person or to different identities. Face matching is crucial for efficient face recognition and plays an important role in applied settings such as passport control and eyewitness memory. However, despite extensive research, the mechanisms that govern face-matching performance are still not well understood. Moreover, to date, many researchers hold on to the belief that match and mismatch conditions are governed by two separate systems, an assumption that likely thwarted the development of a unified model of face matching. The present study outlines a unified unequal variance confidence–similarity signal detection–based model of face-matching performance, one that facilitates the use of receiver operating characteristics (ROC) and confidence–accuracy plots to better understand the relations between match and mismatch conditions, and their relations to factors of confidence and similarity. A binomial feature-matching mechanism is developed to support this signal detection model. The model can account for the presence of both within-identities and between-identities sources of variation in face recognition and explains a myriad of face-matching phenomena, including the match–mismatch dissociation. The model is also capable of generating new predictions concerning the role of confidence and similarity and their intricate relations with accuracy. The new model was tested against six alternative competing models (some postulate discrete rather than continuous representations) in three experiments. Data analyses consisted of hierarchically nested model fitting, ROC curve analyses, and confidence–accuracy plots analyses. All of these provided substantial support in the signal detection–based confidence–similarity model. The model suggests that the accuracy of face-matching performance can be predicted by the degree of similarity/dissimilarity of the depicted faces and the level of confidence in the decision. Moreover, according to the model, confidence and similarity ratings are strongly correlated.

Keywords: face recognition, face matching, unfamiliar faces, signal detection models

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Faces comprise an important class of stimuli in our environment, as they convey a great deal of information regarding such aspects as gender, emotion, and identity (Bruce & Young, 1986; Fitousi & Wenger, 2013). Since the advent of modern photography and digital social networks, the importance of real and photographed faces in our social milieu has increased dramatically (Jenkins et al., 2011; White et al., 2021). A popular laboratory task that has been studied extensively in psychology is the *face-matching task* (Burton et al., 2010; Kemp et al., 1997; Megreya & Burton, 2006, 2007; White et al., 2021). In this task, observers are presented with images of two faces and are asked whether these depict the same personal identity (Burton et al., 2010). Various models of face processing indicate that the perceptual processes involved in face matching are consequential for successful identity recognition (Bruce & Young, 1986). Face matching is also critical in real-world settings, such as passport control and eyewitness identification (Henderson et al., 2001; Wirth & Carbon, 2017; Wixted & Mickes, 2014). Surprisingly, participants are often found to be inefficient in matching identities of unfamiliar faces, and a large body of research has been dedicated to understanding the underlying psychological processes (Bate et al., 2021; Bindemann et al., 2013; Jenkins et al., 2011; Megreya & Bindemann, 2018; Megreya & Burton, 2006; Towler et al., 2014; White et al., 2021).

However, despite extensive research on face matching, the development of a unified model of the task has been thwarted due to two unexpected but robust findings: (a) performance levels in match and mismatch conditions were found to be surprisingly uncorrelated (Megreya & Burton, 2007) and (b) match and mismatch conditions were found to be selectively influenced by different experimental factors (Bate et al., 2021; Fysh & Bindemann, 2023; Kokje et al., 2018; Megreya et al., 2011; Wirth & Carbon, 2017), such as manipulations of specific facial features (Wirth & Carbon, 2017), presentation format (McCaffery & Burton, 2016), or base rates of match trials (Papesh et al., 2018). This apparent dissociation between match and mismatch conditions has led researchers to conclude that performance in the face-matching task cannot be accounted for by a single psychological mechanism (Fysh & Bindemann, 2023). As the present study reveals, these claims turn out to be premature. The present study outlines a unified theoretical framework of the face-matching task—the confidence–similarity unequal variance signal detection (UVSD) model (Egan, 1958) of face matching. This model can explain the face match–mismatch dissociation, along with many other aspects of performance such as the role of confidence and similarity/dissimilarity between the matched faces. The model incorporates basic notions from the classic theory of signal detectability (Green & Swets, 1966; Macmillan & Creelman, 2004) and extends them to account for the unique aspects of face matching. The model posits continuous representations and is contrasted with alternative

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All data, codes, and supporting materials can be downloaded from https://data.mendeley.com/datasets/pntrh5vs8j/2.

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models that postulate all-or-none perception (e.g., high-threshold model). The model is subjected to rigorous empirical testings using hierarchically nested model fitting (Ashby & Perrin, 1988; Rouder et al., 2008), receiver operating characteristics (ROC) curve analyses (O'Toole et al., 2012; Ratcliff et al., 1992; Yonelinas, 1994), and confidence-accuracy plots (Weber & Brewer, 2004; Weber et al., 2013). These testings reveal that the UVSD model outperforms alternative all-or-none models and that it can successfully account for the role of confidence and similarity in face-matching decisions. The model provides novel insights into the basic mechanisms that govern face matching.

The Face-Matching Task

There are two primary experimental procedures for testing face-matching performance. According to one version, participants are presented with a set of multiple study faces (usually 10) along with a target face and are asked whether the target face is present or absent in the study set (Bruce et al., 1999; Megreya & Burton, 2006, 2007). The second version is a same–different task, in which participants are presented with a face pair and are asked to judge whether the two faces depict the same identity or not (Fysh & Bindemann, 2023; Megreya & Burton, 2007; Wirth & Carbon, 2017). The two procedures have been shown to produce comparable results (Megreya & Burton, 2006). What is most remarkable in face matching is that, counter to expectations that humans are experts at face recognition (Diamond & Carey, 1986), people commit relatively high percentage of error in this task. In a landmark research, Bruce et al. (1999) have asked participants to find a target face among 10 lineup face alternatives. The pictures were of excellent quality, but performance was poor, with about 30% error. Similar results were reported in match–mismatch tasks with a pair of faces (Bruce et al., 2001). Moreover, being an expert may not improve performance, as shown in a study by Wirth and Carbon (2017), who tested professional and novices in a passport-matching task.

Many researchers (Fysh & Bindemann, 2023; Megreya et al., 2011; Wirth & Carbon, 2017) assume that a member of a face pair that is easily judged as "match" should also be readily classified as a "mismatch." This led to the prediction of a positive correlation between match and mismatch accuracy rates at the item level. However, contrary to that expectation, the face-matching task has given rise to a dissociation between the two. In a seminal study, Megreya and Burton (2007) have found that accuracy rates for match and mismatch trials with unfamiliar faces were not correlated at the item level. The authors argued that the observed dissociation is at odds with the classic *mirror effect* (Glanzer & Adams, 1985; Glanzer et al., 1993; Osth et al., 2017). In the mirror effect, stimuli that are accurately recognized as "old" are also accurately recognized as "new." Interestingly, familiar faces did show the expected correlation (Fysh & Bindemann, 2023; Megreya et al., 2011; Wirth & Carbon, 2017). A glaring dissociation between match and mismatch conditions has also been documented at the observer level, in the form of weak negative (and not positive) correlations between the two conditions (Bate et al., 2021; Kokje et al., 2018). Moreover, match and mismatch conditions were found to be affected differently by a myriad of factors. For example, Wirth and Carbon (2017) have shown that manipulation of specific facial features impaired performance on mismatch trials by increasing the false-alarm rates, while the rates of correct matching (hits) were not affected. Mismatch accuracy, but not match accuracy was also impaired when one face was presented as a passport picture (McCaffery & Burton, 2016) or when faces were presented under time pressure (Bindemann et al., 2016). In contrast, match accuracy, but not mismatch accuracy, declined when one face in a pair was degraded (Strathie & McNeill, 2016) or the base rates of a "match" were changed (Papesh et al., 2018).

The evidence amassed in support of the dissociation between match and mismatch conditions led researchers to postulate distinct processing mechanisms for match and mismatch conditions with unfamiliar faces. For example, Megreya and Burton (2007) cautioned against the practical implications of this dissociation for forensic psychology: "The results suggest that eyewitnesses' ability to identify a suspect is unrelated to to their ability to reject that suspect" (p. 1183). Similarly, Fysh and Bindemann (2023) concluded that "Together, these findings provide converging evidence that identity-match and -mismatch face pairings are treated as distinct challenges by the cognitive system, and thus require different processes to solve" (p. 5). However, after testing for the role of similarity in face matching, Fysh and Bindemann (2023) admitted that their initial assumption runs into a paradox: "These findings offer somewhat contradictory interpretations. The first of these, based on the accuracy data alone, suggests that match and mismatch trials engage some separable processes. On the other hand, the similarity correlations also imply a common basis for judging whether pairs of faces are identity matches or mismatches" (p. 12).

The similarity data collected by Fysh and Bindemann (2023) cast serious doubts on the idea that face matching is governed by two psychological systems. To understand why, consider the idea that comparison of unfamiliar faces must be an *image-bound* process (Burton, 2013; Jenkins & Burton, 2011), because comparisons are constrained by the visual information within a given face photograph (Hancock et al., 2000; Jenkins & Burton, 2011). Thus, when matching unfamiliar faces, observers must rely on purely perceptual aspects of the given pictures. In this process, it is highly likely that the degree of similarity/dissimilarity between the images may play a central role in the quality of matching. Fysh and Bindemann (2023) tested this conjecture by also asking participants to rate the similarity of the face pairs on a 1–6 scale. They found strict relations between accuracy rates and similarity ratings. For match face pairs, they documented an increase in accuracy rates as similarity ratings increased, while for mismatch pairs, they registered a decline in accuracy rates as similarity ratings increased. These results imply, for example, that high-similarity mismatch pairs are more likely to be classified as "match" than low-similarity mismatch pairs. Moreover, these findings give rise to an apparent paradox. On the one hand, classifications of match and mismatch face pairs are not correlated in terms of accuracy rates, but, on the other hand, match and mismatch face pairs do exhibit strong accuracy–similarity correlations. In other words, one set of findings (accuracy) entails that match and mismatch face pairs are governed by two separate systems, whereas another set of findings (similarity ratings) suggests that match and mismatch face pairs are handled by a single system. How can these apparently opposing theoretical stances be settled?

Additional challenges to the alleged match–mismatch dissociation (Megreya & Burton, 2007) are posed by outcome from research on confidence–accuracy relations (Pleskac & Busemeyer, 2010; Roediger & DeSoto, 2014). If match and mismatch decisions are governed by two separate systems, then a comparable dissociation should emerge in the observed match and mismatch

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confidence–accuracy patterns. Such a dissociation has been documented in the eyewitness memory literature in the form of a *positive–negative asymmetry* (Brewer & Wells, 2006; Sporer et al., 1995; Weber & Brewer, 2004). In this asymmetry, strong linear confidence–accuracy relationships were observed for positive decisions ("this is the person I saw before") and weak or no relations for negative decisions ("this is not the person I saw before"). Surprisingly, such an asymmetry is not found for face-matching decisions. A remarkable study by Stephens et al. (2017) examined this asymmetry along with the effect of proportion of mismatching trials (PMT; Brewer & Wells, 2006) on matching performance. They found that, overall, confidence served as a reliable indicator of accuracy and that for a PMT of 50%, confidence operated symmetrically in match and mismatch decisions. Moreover, when asymmetries did surface in the data, they could be readily removed by equating match and mismatch face pairs on difficulty. These results are important because they reveal that, as far as confidence is considered, match and mismatch decisions are not governed by two separate mechanisms.

In summary, the claims for two separate psychological systems of face matching might be premature. First, there may exist a theoretical framework that can explain the empirical results in terms of a single system. Second, the interplay between effects of perception and decision, that can explain many of the empirical results, is not taken into consideration. Third, important aspects of decision making and judgment such as *confidence* are not taken into account. Fourth, when similarity data are considered, the idea of two systems runs into a paradox. All of these suggest that a unified theoretical framework of face matching is highly needed. The current effort proposes such a framework.

The Present Study

In the following sections, I present a UVSD model (Wixted, 2007) of face matching. Although the model focuses on a single task (i.e., matching), it offers a more general view of face recognition, one that takes into consideration the contribution of both "between-identities" and "within-identities" variability (Jenkins et al., 2011). The great majority of face-space models (Nosofsky, 1988; Trujillo & Anderson, 2023; Valentine, 1991; Vanpaemel & Storms, 2008) are not designed to cope with variability in the photos of the same identity. This theoretical lacuna has likely impeded progress in face research (Burton, 2013; Kramer et al., 2018). Thus, the current approach also develops a more general (featural) framework to face recognition that is capable of representing both between-identities and within-identities variabilities. Before I present the model and related general frameworks in detail, it is important to distinguish between these two sources of variability in face processing.

"Within-Identities" and "Between-Identities" Sources of Variation

Our ability to recognize images of faces is governed by two sources of variability. First, variability caused by the existence of numerous facial identities. Second, variability that results from the presence of many photos (or real-world appearances) of the same person that do not appear to us in the same exact form or context due to changes in photo (e.g., illumination) or in the person (e.g., age or expression). A first step toward developing a comprehensive account of face matching is to distinguish between these two factors: (a) *within-identities* variation, which underscores changes within a given facial identity due to aging, position, camera, lighting, and so forth (Burton et al., 2011, 2016; Jenkins et al., 2011), and (b) *between-identities* variation, which reflects the existence of different facial identities (Kramer et al., 2018; Moses et al., 1994; White et al., 2022). Within-identities and between-identities sources of variation likely affect the way we perceive and match faces or even learn new identities (Kramer et al., 2017). The former type of variability may compromise our ability to defeat the second type of variability and generalize over different photos to retain a coherent and invariable representation of identities (Jenkins et al., 2011).

The role of ambient noise in face recognition gained traction in various disciplines, including psychology (Jenkins et al., 2011), neuroscience (Freiwald et al., 2009), and computer vision (Adjabi et al., 2020; Huang et al., 2008; Moses et al., 1994). Notably, photos of the same facial identity can vary more than images of different facial identities and, consequently, hamper our ability to recognize faces. This has been demonstrated in a seminal study by Jenkins et al. (2011). Participants were presented with a deck of ambient photos of two unfamiliar identities and asked to categorize these images to as-many-facial-identities as they thought were present in that deck. Remarkably, participants partitioned the deck into an average number of seven facial identities! The participants did not realize that there were only two facial identities in the deck, giving currency to the notion that within-identities variation can sometimes defeat between-identities variation (White et al., 2022).

Variability in photos of the same facial identity poses a serious challenge not only for humans but also for automatic systems. In an early study, Moses et al. (1994) created a large database of photos by photographing 26 different identities under various conditions of illumination, viewing position, and expression. Several measures of distances between the two dimensional image representations of the images were computed. The authors found that variability due to illumination and viewing directions was larger than variability due to the presence of various identities. The authors tested the ability of an automated system to classify those images based on a measure of distance between two images. When the direction of illumination changed from left to right, the system failed to recognize all the images in the database (100% missed faces). This level of performance improved a bit to 80% missed faces only when the representations were sensitive to horizontal features (e.g., sine Gabor filter at orientation of 90°). Similar results were obtained with respect to viewing position. With regard to expression, the system performed well when the variability was due to a smile (0% missed faces) but deteriorated again when only parts of the face (e.g., eyes area) were presented. Since Moses et al. (1994) study, almost 30 years have passed and, obviously, considerable progress has been made in the capabilities of automatic systems to recognize faces (Huang et al., 2008). However, limitations due to "within-identities" variability are still threatening their validity, as noted in a recent comprehensive review by Adjabi et al. (2020): "The intra-class variations are larger than the inter-class variations, contributing to low results in recognition" (p. 39).

The two sources of variability in face images are crucial for understanding how we learn new faces (Kramer et al., 2017) and, particularly, what are the transformations that novel faces undergo in our brain on their way to becoming familiar (Kramer et al., 2018). To examine this question, White et al. (2022) employed an analysis of variance (ANOVA)-like approach as a means of delineating the

contributions of the two sources. Participants were asked to rate the similarity of gender averages, which served as a proxy for the centroid of face space, to an identity average that represented the appearances of a single identity. These ratings served to measure between-identities variability. In addition, participants rated the similarity of the identity averages to exemplar images that were used to compute these averages. These ratings gauged within-identities variability. The authors conjectured that the degree of familiarity of faces should affect the within-identities component. Indeed, images of a given familiar identity appeared as more similar to the average appearance of that identity than those of an unfamiliar identity, and this modulation was stronger than that observed for between-identities ratings. Moreover, reduction in within-identities variation was more pronounced in a group of super recognizers than in typical viewers.

Based on the reviewed studies, three main conjectures can be made. The first concerns within-identities and between-identities sources of variance. In standard tests of face matching (Burton et al., 2010; Fysh & Bindemann, 2018), where the experimenters compose their face pairs according to their own criteria, different relations may arise between these two sources of variability. For example, to render their face-matching test more challenging, Rice et al. (2013) have purposely chosen match face pairs that were rated with low-similarity scores and mismatch face pairs that were rated with high-similarity scores. This deliberate intervention with the stimuli resulted in lower levels of accuracy, likely due to alteration of within-identities and between-identities variances. Second, perceived similarity/dissimilarity is a viable metric for measuring variability and is likely the variable according to which observers (and automatic systems) make their decisions. Third, experimental factors that affect performance with ambient faces are likely operating on either within-identities and/or between-identities variability. To model the continuous representation of similarity along with the two sources of variation in face perception, a UVSD model is proposed.

An Unequal Variance Signal Detection Model of the Face-Matching Task

The UVSD (Egan, 1958) has been used extensively in psychology, mainly in modeling old/new recognition memory decisions (Wixted, 2007; Wixted & Stretch, 2004). Recently, the model has been successfully applied to eyewitness identification of faces (Colloff et al., 2021; Gronlund et al., 2014; Shen et al., 2023; Wixted, 2020; Wixted & Mickes, 2014). It is quite surprising that, to date, a full-blown signal detection theory (SDT) approach to face matching has not been developed. Few researchers did employ signal detection measures in face-matching experiments, but only for descriptive purposes (Howard et al., 2020; Menon et al., 2015; Papesh et al., 2018; Stephens et al., 2017; Wirth & Carbon, 2017). Heretofore, a complete theoretically motivated SDT approach that takes into consideration similarity relations, confidence, and the distinction between within-identities and between-identities variability, as well as their unique contributions, has not been advanced. The current UVSD model offers such an approach. Here, I demonstrate how the UVSD can be successfully applied to face matching.

The novel application of the UVSD is based on the following assumptions: (a) On each trial, a given face pair evokes a sensation in the observer that is associated with the strength of an identity signal. This identity signal conveys information regarding the degree of similarity/dissimilarity of the two faces, and in this sense, it is agnostic with respect to the sensation elicited by the individual faces that comprise the face pair. The more similar the faces, the stronger is the identity signal; (b) both match and mismatch face pairs elicit identity-strength signals that vary in their strength; (c) sensation (identity-strength signal) is assumed to be a random variable because the strength of the signal changes from trial to trial in a random fashion; (d) identity-strength signals produced by match and mismatch face pairs are generated, respectively, by *within-identities* and *between-identities* sources of variation. These are represented in the model by two overlapping Gaussian distributions (i.e., $\mu_{match} > \mu_{mismatch}$), with unequal variances (i.e., $\sigma_{match} \neq \sigma_{mismatch}$), the match distribution represents within-identities variation, whereas the mismatch distribution reflects between-identities variation; (e) the observer places a decision criterion on these distributions; and lastly (f) the combination of the identity-strength continuum and the location of the response criterion gives rise to both calling the face pair "match" (or "mismatch") and to acquiring a certain level of confidence in the decision (Roediger & DeSoto, 2014). The observer is using a response rule: If the strength of the identity signal is greater than the criterion, then the face pair is a "match," otherwise the face pair is a "mismatch." One implication of this model is that confidence is derived by *experienced similarity* between the faces that compose the presented face pair. The model suggests that confidence is high when the faces are highly similar or very dissimilar and that confidence is low when similarity is intermediate.

The evidence adduced by behavioral (Jenkins et al., 2011; White et al., 2022) and computer vision (Adjabi et al., 2020; Huang et al., 2008; Moses et al., 1994) studies suggests that variabilty in photos of the same identity (match) can sometimes be higher than variabilty in photos of different identities (mismatch). Notably, Jenkins et al. (2011) have demonstrated how participants classify a deck of images that convey only two identities to (on average) seven different identities. Based on this evidence, one would predict that the variance of the match distribution in the UVSD is expected to be higher than the variance of the mismatch distribution. However, as noted earlier, it is possible that in standardized face-matching tests, such as the Glasgow Face-Matching Test (GFMT; Burton et al., 2010) or the Kent Face-Matching Test (KFMT; Fysh & Bindemann, 2018), this assumption may be violated. This is because experimenters deliberately or unknowingly intervene with the psychological structure of their stimuli set. For example, by equating the difficulty of match and mismatch pairs or by pairing faces in a nonrandom fashion. In the study by Rice et al. (2013), for instance, the researchers deliberately constructed "match pairs" only from highly dissimilar images, and "mismatch pairs" only from highly similar images. Their aim was to increase the difficult of the test. This practice likely selectively influenced the location and variance of match and mismatch distributions. Moreover, it is theoretically plausible that the relations between the variances of the two distributions change as a function of experimental manipulations that selectively influence one of them. A case in point is alteration of specific facial features in the study by Wirth and Carbon (2017), which resulted in an increase in false-alarm, but not hit rates. In the next sections, I outline a feature-matching model that can explain how such manipulations may affect the variances of the pertinent distributions. Other effects of theoretical importance are related to familiarization (Burton et al., 2016). According to this idea, as participants become more familiar with images of the same identity, these images appear more similar and as a result within-identities

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variance (match distribution) decreases; in contrast, the similarity between images of different identities decreases and as a result between-identities variance increases. This is exactly what Kramer et al. (2018) have found in a computer vision study. Such a process can explain why matching familiar faces, in contrast to unfamiliar faces, is considered to be a highly accurate task (Megreya & Burton, 2006). The featural face space I outline next can account for these effects.

Thus, the current UVSD model assumes that natural settings with random pairings of face images will result in higher variance in the match compared to mismatch distributions. However, in standardized face-matching tests, the relations between the variances are task-dependent because the construction of the stimuli set is under the experimenter's direct control, and this determines how similar/dissimilar those faces appear to the observer. Given that similarity is the major factor in face matching, within-identities and between-identities similarities in such tests dictate the variances of the match and mismatch distributions. The upshot is that with respect to face-matching canonical tests, the prediction on which distribution will have a higher variability depends on the similarity relations established by the experimenter. It is interesting to note that such similarity relations are also critical for performance in eyewitness lineup procedures (Wixted & Wells, 2017). It turns out that the outcome of eyewitness tests is affected by how filler faces are chosen based on their similarity relations to the suspect. Colloff et al. (2021) have shown that selecting fillers that are judged as dissimilar to the suspect inflated the hit rate, but did not affect the false-alarm rate. These authors could reverse this effect by matching the fillers' similarity to the suspect in both target-present and target-absent lineups. These results are intriguing because they give currency to the notion that the experimenter's intervention with the similarity relations of her stimuli can induce selective influences on the match and mismatch distributions. This is exactly what the current UVSD assumes. A featural face-space model will be developed soon to demonstrate how these relations are determined.

The UVSD model can be readily applied to face-matching tasks with multiple faces (Megreya & Burton, 2006). In that case, the identity signal is produced for face ensembles rather than face pairs, and the logic of the UVSD is principally the same. In any event, it is important to note that the identity signal reflects the degree of similarity/dissimilarity of the two faces and is blind to the properties of the individual faces that compose a face pair. The best metaphor for this process is that of a *t* test for paired samples, in which the statistical inference is held with respect to the difference of means rather than the means themselves.

The UVSD model is composed of three parameters: a sensitivity parameter d', which reflects the distance between the match and mismatch distributions; a response bias parameter c, which designates the locus of decision criterion set by the observer; and a standard deviation parameter σ, which reflects the width of the match distribution. The standard deviation of mismatch distribution is set to 1. The probabilities of hit and false alarm in this model are as follows:

$$p_h = 1 - \Phi\left(\frac{c - d'}{\sigma}\right)$$
$$p_f = 1 - \Phi(c). \quad (1)$$

The UVSD model is based on the assumption that face perception is affected by both within-identities and between-identities sources of variability (White et al., 2022). Another assumption is that with unfamiliar faces, the standard deviation of the match distribution is expected to be larger than the standard deviation of the mismatch distribution. This assumption is based on behavioral (White et al., 2022) and computer vision (Moses et al., 1994) studies. It should be noted though that in the context of an experiment, the relations between the two sources of variability can be deliberately or unknowingly altered by the experimenter. For example, by pairing same-identity images that are more similar to each other, within-identities variability can become lower than between-identities variabilty. This may occur in standard face-matching tests (Burton et al., 2010; Fysh & Bindemann, 2018), when, for example, experimenters strive to equate the difficulty of match and mismatch conditions.

The strongest support for the use of SDT in psychology came from the analysis of ROC curves (Cook & Wixted, 1997; Macmillan & Creelman, 2004; Mickes et al., 2017; Ratcliff et al., 1992; Wixted & Mickes, 2014; Yonelinas, 1994; Yonelinas & Parks, 2007).

ROC Curves

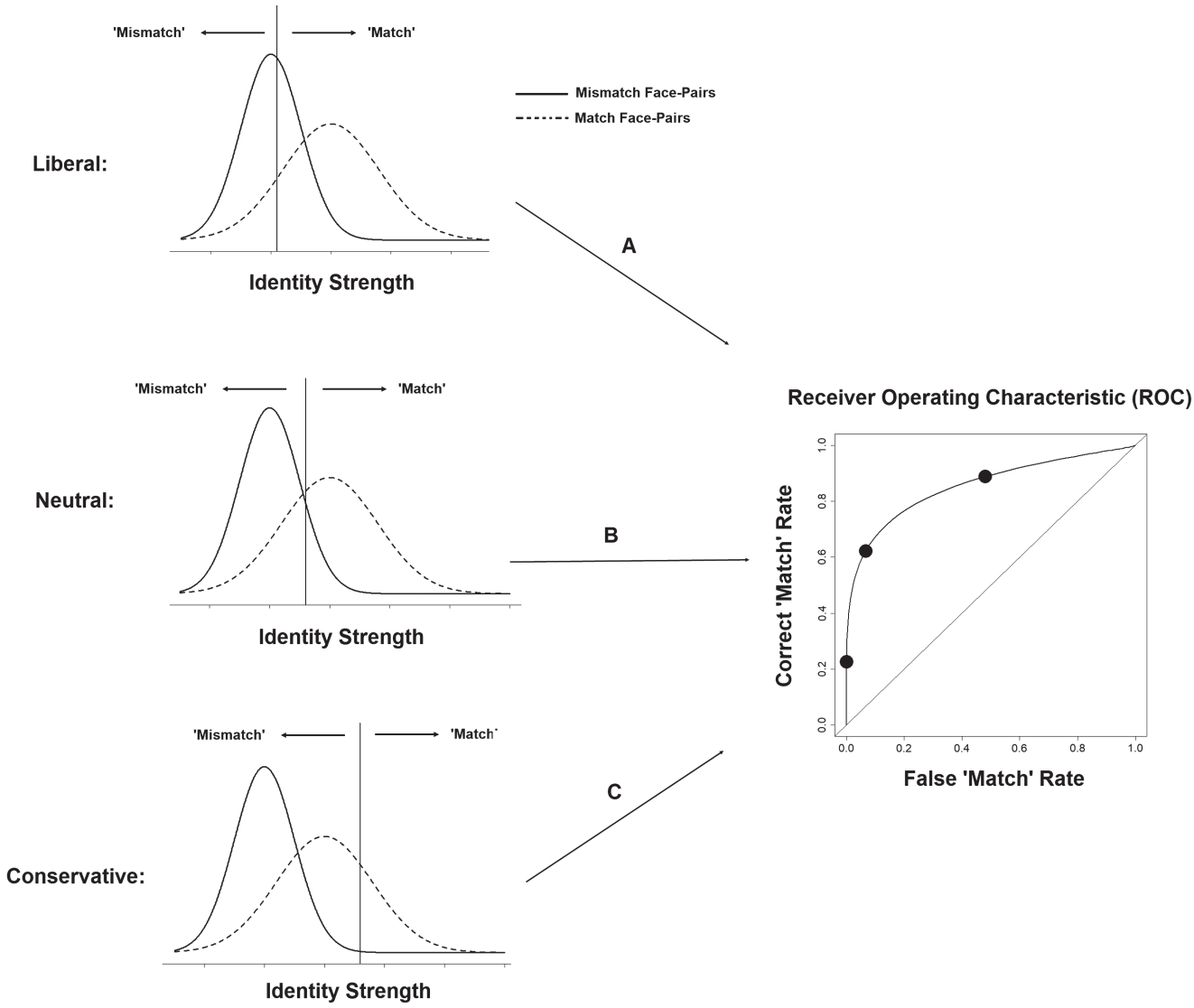
An ROC is a function that relates the rate of hit rates to the rate of false alarms (Green & Swets, 1966; Macmillan & Creelman, 2004). These points are affected by discriminability and response bias. Discriminability is reflected by the overlap between the match and mismatch distributions. The more the mismatch and match distributions overlap, the less able observers are to discriminate between match and mismatch face pairs. In contrast, response bias refers to the placement of the decision bound on these distributions. It is under the control of the observer (but see Miller & Kantner, 2020) and can be manipulated by various means such as by instructions. For example, the recognition of a match can be emphasized, and as result, the observer will adopt a liberal criterion, which will produce high rates of hits but also high rates of false alarms. A conservative criterion, on the other hand, will result in lower rates of hits, but also of false alarms. Figure 1 presents three response criteria along with their corresponding placements on an ROC curve. As can be noted, as the response criteria become more conservative, both the hit and false-alarm rates decrease. The points on the ROC create an isosensitivity line. The shape of the ROC conveys important information and ROCs can be used to dissociate between competing models of SDT (Mickes et al., 2017; Wixted & Mickes, 2014; Yonelinas, 1994).

A Unified Confidence–Similarity UVSD Model of Face Matching

Panel A in Figure 2 illustrates the UVSD model of face matching as it is tested in a confidence-rating task. Confidence ratings are often used to generate the ROC curves (Macmillan & Creelman, 2004). The UVSD can be readily adapted for confidence-rating data in the face-matching task. On each trial, the participant is presented with a face pair and is asked to judge whether the faces depict the same identity or not. The participant gives her response by indicating whether the faces form a "match" or a "mismatch" and whether their decision is given with "high," "medium," or "low" confidence. Practically, the participant has to choose from the following labels: 1 = *mismatch high confidence*, 2 = *mismatch*

Figure 1

An Illustration of the Unequal Variance Signal Detection Model as It Is Applied in the Current Work to Face Matching



Note. The model consists of two distributions: one for mismatch face pairs and another for match face pairs. The three panels depict three levels of response criteria: (A) liberal, (B) neutral, and (C) conservative. In all three panels, the mean and standard deviation are 0 and 1, respectively, for the mismatch distributions and 1 and 1.6, respectively, for the match distributions. The receiver operating characteristic (ROC) to the right illustrates the three ROC points corresponding to the three response criteria setting presented on the left. The ROC points "A," "B," "C" correspond, respectively, to the liberal, neutral, and conservative models depicted on the left-hand side.

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medium confidence, 3 = *mismatch low confidence*, 4 = *match low confidence*, 5 = *match medium confidence*, and 6 = *match high confidence*. In the general case, with N degrees of confidence, the data can be modeled as generated by the observer placing $N - 1$ decision bounds $c_1, c_2 \ldots c_{N-1}$, such that $c_1 \leq c_2 \leq \ldots c_{N-1}$. In the current confidence-rating framework, six degrees of confidence yield five response criteria. Panel A of Figure 2 illustrates such a model. Identity strength below c_1 produces the response "face do not match with high confidence," identity strength above c_1 and below c_2 results in the response "face do not match with medium confidence," and so forth. The UVSD has seven parameters (d, σ, c_1, c_2, c_3, c_4, and c_5) and it predicts curvilinear ROC curve (see Panel B of Figure 2).

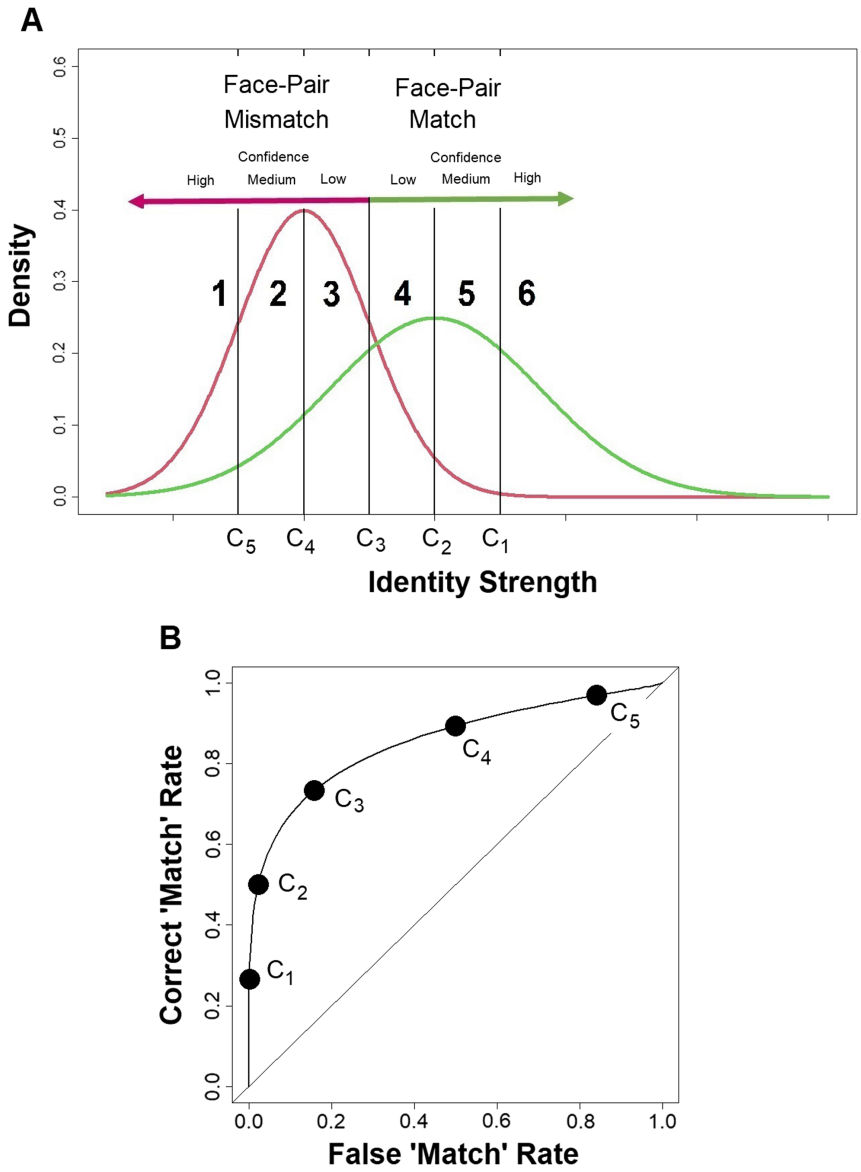
Predictions of the Unified Confidence–Similarity UVSD Model

Face-Matching Phenomena

The UVSD can account for basic findings in the face-matching literature, such as the presence of individual differences in matching ability (Bate et al., 2021; Smith et al., 2021; White et al., 2014;

Figure 2

An Unequal Variance SDT Model of Face Matching



Note. Panel (A) depiction of the UVSD model of face matching with the six levels of confidence ratings and their mapping onto response areas on the match and mismatch distributions. Confidence ratings are numbered 1–6, such that 1 = *mismatch high confidence*, 2 = *mismatch medium confidence*, 3 = *mismatch low confidence*, 4 = *match low confidence*, 5 = *match medium confidence*, and 6 = *match high confidence*. Confidence ratings are associated with response criteria C_1–C_5. Identity-strength values above C_3 give rise to a "match" decision, whereas identity-strength values below C_3 elicit a "mismatch" decision. Panel (B) depicts the corresponding ROC curve for the confidence rating UVSD model. Points C_1–C_5 correspond to response criteria C_1–C_5 in Panel A. UVSD = unequal variance signal detection; ROC = receiver operating characteristics; SDT = signal detection theory. See the online article for the color version of this figure.

Wirth & Carbon, 2017), or differences in overall accuracy across different face-matching tests (Burton et al., 2010). Take for example, two of the most popular face-matching tests, the GFMT (Burton et al., 2010) and the KFMT (Fysh & Bindemann, 2018). The two tests produce highly dissimilar accuracy rates, such that the former produces much higher accuracy rates than the latter. These differences can be readily modeled as differences in discriminability in match and mismatch face pairs across the two tests. The UVSD can explain more puzzling key aspects of face-matching performance, such as the finding that different experimental factors exert selective

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influence on match (mismatch) but not mismatch (match) conditions (Megreya & Burton, 2006, 2007). The UVSD can readily account for this phenomenon by adhering to the unequal variances of the match and mismatch distributions. In particular, some experimental factors may affect only one of the distributions (e.g., match), increasing or decreasing its standard deviation, while leaving the other (e.g., mismatch) intact. Take for example, the finding that manipulation of specific facial features hampered performance on mismatch trials by increasing false alarms, but did not change hit rates (Wirth & Carbon, 2017). A plausible explanation is that the manipulation increased the variance of the mismatch distribution but left the variance of the match distribution unaffected. In the next section, I propose a feature-matching mechanism of similarity computation that can explain how different manipulations alter the variability of the distributions.

The UVSD can also explain the allegedly puzzling observation that accuracy rates in match and mismatch conditions are not positively correlated at the item level (Fysh & Bindemann, 2023; Megreya & Burton, 2006, 2007; Wirth & Carbon, 2017). The application of the UVSD to face matching is based on the assumption that observers engage in discriminating between *face pairs* and not between *individual faces*. Accordingly, an identity-strength signal is generated by a face pair not by a single individual item. Thus, the identity signal of a particular item face depends on its pairing with other item faces and therefore cannot be separated from the context in which it was presented. The same item (face) can give rise to different (uncorrelated) identity signals, depending on its pairing condition (match or mismatch) with other faces. An easy-to-recognize face in a match pair may not be an easy-to-recognize item when positioned next to another item in a mismatch pair. This implies that in this model, the rates of hit and false alarm for a given item are not expected to be correlated.

This formulation of the UVSD model also offers a mechanism for understanding how the internal representation of unfamiliar faces changes as a function of learning (Burton et al., 2016; Kramer et al., 2017; Ritchie & Burton, 2017; Ritchie et al., 2021; Sandford & Ritchie, 2021). The mechanism is based on reduction in the within-identities variability, namely in the standard deviation of the match distribution. In this respect, the model bears some resemblance to the ANOVA approach advocated by White et al. (2022), according to which the accuracy of correct categorization of a given image is determined by the ratio of between-identities and within-identities variability of the images in the experimental set. As this ratio becomes larger, the evidence for a true difference between the two distributions increases. Thus, this model is also a model of perceptual learning. As faces become more familiar, the degree of within-identities similarity is increased and the variability of the representation decreases. This can explain why images of highly familiar faces are not subjected to the same confusion as unfamiliar faces. Several researchers noted that SDT models cannot be applied to familiar faces because performance with these stimuli is at ceiling (Megreya & Burton, 2007) claiming that "unfamiliar faces are not faces" (Megreya & Burton, 2006). The proposed UVSD model puts unfamiliar and familiar on equal footing by offering a learning mechanism of reduction in variability (and increase in perceived similarity) in same-identity photos.

An important question that arises is how the match and mismatch distributions in the UVSD are generated and what factors determine their spread. The signal that is elicited by each face pair is likely determined by the faces' representations in a more general face space (Valentine, 2005). But as noted earlier, extant face-space models cannot adequately represent variation in the photos of the same identity. Therefore, the next section develops a featural face space that is capable of representing and generating the within-identities and between-identities variability in the UVSD distributions.

Modeling Similarity in Face Matching

As noted earlier, face matching with unfamiliar faces is considered by many researchers as an *image-based* process (Hancock et al., 2000; Jenkins & Burton, 2011). According to this idea, unfamiliar faces are compared based on their pictorial information. Similarity is a fundamental explanatory construct in models of face perception (Colloff et al., 2021; Shen et al., 2023; Valentine, 1991). It is therefore highly likely that the resemblance between the depicted images operates as a mediating variable of face matching. In many face-modeling approaches, similarity is the driving force behind robust phenomena such as distinctiveness or other-race effects (Trujillo & Anderson, 2023). In canonical face spaces (Valentine, 2005), similarity is construed as a distance in some Euclidean space. Similar faces are located more closely in that space than dissimilar faces. Similarity has been found to affect face-matching performance of both automatic systems and humans. For example, an application of the principal component analysis (PCA) to stimuli from the GFMT by Kramer et al. (2018) has revealed that the average Euclidean distance for "match" pairs was smaller than for "mismatch" face pairs. The role of similarity in human face matching has been recently investigated by Fysh and Bindemann (2023), who demonstrated that strong lawful relations ensue between accuracy rates and similarity ratings in the face-matching task. In their study, the degree of similarity/dissimilarity between the images in face pairs predicted the pattern of responses to those face pairs both at the global face level and at the featural level.

Similarity plays a key role in the current UVSD model as well. The decision variable in the model is an identity-match signal that is based on the perceived degree of similarity between the two faces. Modeling similarity is therefore quintessential in accounting for face-matching performance. Similarity has been shown to play an important role in eyewitness identification (Brewer & Wells, 2006; Colloff et al., 2021). A comparable UVSD model to the one presented here has been recently proposed by Shen et al. (2023). In their model, the memory-match signal is assumed to reflect the degree of similarity between two faces (a face in the lineup and the face of the perpetrator in memory), and similarity is modeled by a feature-matching process. In the current UVSD model, the decision variable is an identity signal, which relies on the degree of similarity/dissimilarity between the two faces. In the UVSD model, match and mismatch distributions are allowed to differ in their variance, reflecting differences in the degree of similarity/dissimilarity between match and mismatch face pairs. In the next section, I will outline a feature-based mechanism of similarity computation that can account for the unequal variability in the match and mismatch distributions.

Before I outline the proposed mechanism of similarity computation, a word is in order regarding the concept. Similarity is an essential explanatory construct in psychological theorizing (Nosofsky, 1986); but it is also notorious for being a slippery concept (Algom & Fitousi, 2016; Melara, 1992). The popular idea that similarity of two objects can be modeled as the Euclidean distance between two points in multidimensional space (Shepard, 1962) has been famously criticized by Tversky (1977), who presented convincing demonstrations that

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similarity judgments violate central mathematical axioms of distance such as minimality (i.e., A should resemble A more than it resembles B) and *symmetry* (i.e., A should resemble B to the same degree that B resembles A). Violations of such fundamental distance axioms pose serious challenges also to canonical face recognition models and in particular to face-matching theories. Judging an image of a given identity as more similar to that of another identity (Jenkins et al., 2011) implies a violation of the minimality axiom (also termed "identity axiom"). Thus, when classic approaches to face spaces, such as Multidimensional Scaling (Shepard, 1962) or norm-based models (Valentine, 1991), are applied to ambient images of faces that are affected by within-identities variability, they run into conceptual problems (see also White et al., 2022). Feature-based models can overcome these difficulties. Tversky (1977) proposed a feature-based model that quantifies similarity as the degree to which objects have common and separate features.

$$S(b,a) = \theta f(B \cap A) - \alpha f(B - A) - \beta f(A - B), \tag{2}$$

where A and B are feature sets representing Face a and Face b, respectively. f is some measure over feature sets, and θ, α, and β are free parameters of the model. Similarity involves a contrast between the common features of B and A, ($B \cap A$), and their distinctive features, those owned by B but not A ($B - A$), and those possessed by A but not B ($A - B$). When $\alpha = \beta = 0$, the model is reduced to a feature-matching model in which similarity reflects the common features. Next, I present a feature-matching model (Shen et al., 2023) that is capable of generating UVSD representations.

A Feature-Matching Similarity Model

Many cognitive (Abudarham et al., 2019; Colloff et al., 2021; Ellison & Massaro, 1997; Shen et al., 2023; Tversky & Krantz, 1969; Wixted & Mickes, 2014) and biological (Freiwald et al., 2009) models of face perception are based on the premise that feature analysis is necessary for the computation of similarity and other quantitative aspects of face recognition. According to these models, each face is a point or a finite feature description in a vector space representation. The goal of the current modeling approach is to demonstrate how a face-matching operation that is based on such discrete feature vector space representation can generate a binomial signal detection model that can be approximated by a Gaussian SDT model. I also demonstrate how such an operation, when held with same-identity and different-identity trials, can lead to two separate binomial or Gaussian distributions characterized by different values of mean and variance. The binomial signal detection model has precedent in the memory literature (Glanzer & Adams, 1990; Glanzer et al., 1993). To account for the mirror effect in memory (Glanzer & Adams, 1985), Glanzer and his colleagues have developed a binomial SDT model in which participants attend to featural information and use a likelihood-ratio decision rule. The current feature-matching model offers a similar mechanistic account of the unified similarity–confidence UVSD face-matching model. All of these are accomplished by adhering to rudimentary aspects of a feature-by-feature discrete-matching operation.

First, I assume that a face is represented by n features ($f_1 \rightarrow f_n$), with each feature taking one of v possible settings (Shen et al., 2023). The term "features" here is applied in a broad sense (see Shen et al., 2023) and can include, in addition to obvious featural aspects such as eyes shape (e.g., round, wide) or nose shape (e.g., straight, convex), also configural relations (e.g., eye–nose ratio), as well as other aspects such as race (e.g., Caucasian, Asian). In the face-matching task, Faces A and B are represented as two vectors of feature settings $\vec{a} = (a_1 \ldots a_n)$ and $\vec{b} = (b_1 \ldots b_n)$. When a face pair is presented, the observer independently encodes the features from the two faces in a lossless compression. Then the observer performs a series of independent exhaustive comparisons between the two vectors across all features (Bradshaw & Wallacei, 1971). Exhaustive processing (Fitousi, 2021b; Sternberg, 1966; Townsend & Ashby, 1983) of facial features is a reasonable assumption given their allegedly holistic nature (Farah et al., 1998). Several studies (Wenger & Townsend, 2001, 2006) have documented exhaustive stopping rule with facial features (Fitousi, 2021b). Next, the outcome of each feature comparison is represented as either 0 (= mismatch) or 1 (= match), which yields a vector of zeros and ones $\vec{z} = (z_1 \ldots z_n)$. Similarity is computed as the total number of matches summarized over all features in $\vec{z}$:

$$S(ab) = f(A \cap B) = \sum_{i=1}^{n} z_i, \tag{3}$$

where $S(ab)$ is the similarity value, and $A \cap B$ is the subset of common features (Tversky, 1977). The observer emits a "match" response only if similarity is perfect, that is, all features elicited a match signal and therefore $S_{ab} = n$; otherwise, the observer responds with a "mismatch" decision. Other decision rules can also be modeled based on the similarity signal. For example, a "match" response can be made even when the similarity score is not perfect, but only exceeds a given value (e.g., 9 features match out of 10). But for the sake of simplicity, this possibility is not pursued here. This similarity signal is analogous to the identity signal assumed in the face-matching UVSD model.

To illustrate the model, consider the following numerical example. A face-matching task is accomplished by representing five features ($f_1 \rightarrow f_5$), with each feature receiving one of four possible settings ($v = 1, 2, 3, 4$): f_1 = lips (1 = wide, 2 = thin, 3 = round, and 4 = full); f_2 = eyes (1 = round, 2 = wide, 3 = downturned, and 4 = upturned); f_3 = nose (1 = straight, 2 = convex, 3 = concave, and 4 = wavy); f_4 = forehead (1 = broad, 2 = narrow, 3 = curved, and 4 = sharp); and last, f_5 = ear (1 = pointed, 2 = square, 3 = broad lobe, and 4 = narrow). In this example, Face A and Face B are represented as vectors $\vec{a} = (2,1,4,3,3)$ and $\vec{b} = (2,3,4,1,4)$. The matching operation results in a vector $\vec{z} = (1,0,1,0,0)$, which describes the outcome of this feature-by-feature matching operation as a series of matches (1) and mismatches (0). Similarity is computed as the sum of matches over all features $S_{ab} = f(A \cap B) = \sum_{i=1}^{5} z_i = 1 + 0 + 1 + 0 + 0 = 2$. It is assumed that the feature matching operation takes the form of a Bernoulli process, with a match rate of p and a mismatch rate of $1 - p$. The similarity signal is distributed as the probability of k feature matches out of n features. This can be captured by the binomial distribution:

$$f(k,n,p) = \Pr(k;\, n,p) = \Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}. \tag{4}$$

A global "match" decision is emitted only when all features elicit a "match" signal ($k = n$) and similarity is maximal ($S_{ab} = n$). The expected value of the distribution is $E[X] = np$. Notably, $E(X)$ is the

sum of n features with expected value of p and therefore can be written as $E(x) = E[X_1 + \ldots + X_n]$. This formal aspect is consistent with the idea that each feature contributes an independent share to the overall similarity (see Fysh & Bindemann, 2023). In addition, the variance of the distribution is $\mathrm{Var}(X) = np(1-p) = np - np^2$. The variance of the sum of independent random variables is the sum of the variances. This formality is also commensurate with the idea that each facial feature contributes an independent component of variance to the overall representation. Most importantly, if n is large enough, the binomial distribution can be approximated by a Gaussian distribution.

$$x \sim \mathcal{N}(\mu = np, \sigma^2 = np - np^2). \tag{5}$$

The mean of the distribution is equal to the product of the number of features n and the feature match rate p, and its variability amounts to $np - np^2$.

The Gaussian distribution represents a continuous similarity signal that is produced by a discrete feature-by-feature matching operation. This is a simple mechanism that transforms a series of discrete match/mismatch decisions with respect to individual features onto a continuous global similarity signal in the form of a Gaussian distribution. In this model, both the mean (μ) and variance (σ^2) of the distribution are determined by two parameters: the number of features compared n and the rate of feature matching p. We can assume that the feature-matching rate p is different in match and mismatch trials, while the number of compared features n is held fixed across trials. This leads to a model with three parameters that generate two Gaussian similarity signal distributions, one for match trials:

$$G_{\text{match}} \sim \mathcal{N}(\mu_{\text{match}} = np_1, \sigma^2_{\text{match}} = np_1 - np_1^2), \tag{6}$$

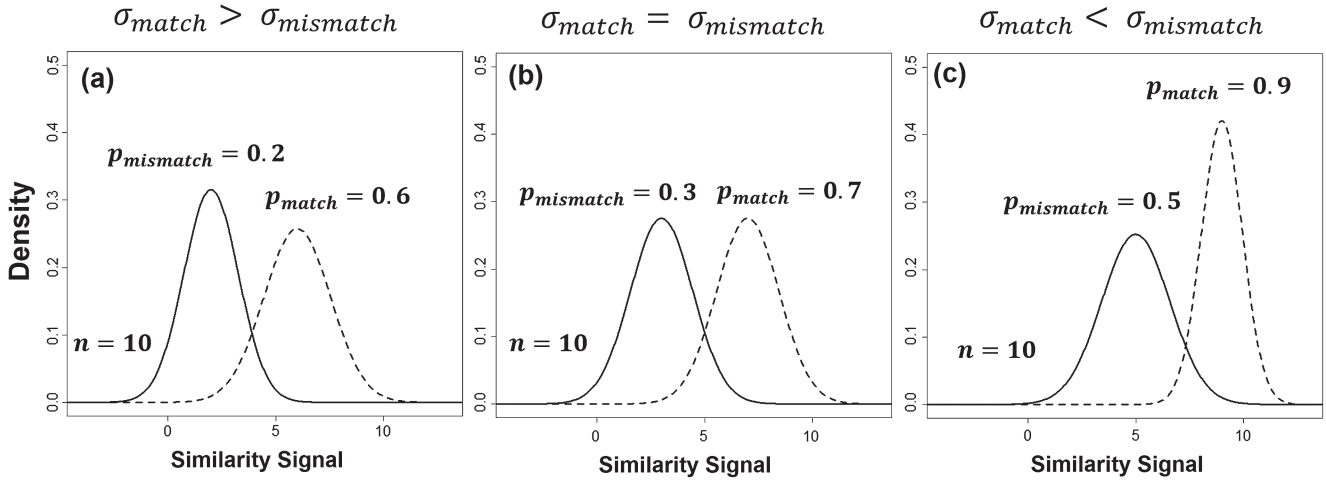
and another for mismatch trials:

$$G_{\text{mismatch}} \sim \mathcal{N}(\mu_{\text{mismatch}} = np_2, \sigma^2_{\text{mismatch}} = np_2 - np_2^2), \tag{7}$$

where p_1 is the feature-matching rate for match trials and p_2 is the feature-matching rate for the mismatch trials. The ratio of variances amounts to $\frac{\sigma^2_{\text{mismatch}}}{\sigma^2_{\text{match}}} = \frac{np_2 - np_2^2}{np_1 - np_1^2}$, and discriminability can be computed by $d' = \mu_{\text{match}} - \mu_{\text{mismatch}} = np_1 - np_2$. One notable aspect of the model is that the discriminability and variances are not independent entities. In addition, for the representations to be consistent with major findings in face-matching studies, the mean signal of the match distribution should be greater than the mean signal of the mismatch distribution ($np_1 > np_2$, which implies that $p_1 > p_2$). These values are also subjected to the constraints that $np_1 - np_1^2 > 0$ and $np_2 - np_2^2 > 0$. The feature-matching rate p_1 is often higher in match (i.e., within-identities) trials than in mismatch p_2 (i.e., between-identities) trials, a difference that generates the unequal variances of the distributions. The direction of the inequality is determined by the specific values taken by parameters p_1 and p_2. As for the n, the parameter that reflects the number of matching features, this variable is likely kept under the experimenter control and thus remains constant along a given experiment. However, it can change from experiment to experiment due to specific settings (e.g., one experiment uses only internal facial features, whereas a different study also includes external facial features like hair and ears).

A numerical example can demonstrate how this model works. Assume that there are 10 features ($n = 10$) and that the feature-matching rate in match trials is $p_1 = 0.6$ and in mismatch trials $p_2 = 0.2$. This entails stronger mean similarity signal in the match distribution ($\mu_{\text{match}} = 10 \times 0.6 = 6$) than in the mismatch distribution ($\mu_{\text{mismatch}} = 10 \times 0.2 = 2$) and higher variability in the match distribution ($\sigma^2_{\text{match}} = 10 \times 0.6 - 10 \times 0.6^2 = 2.4$) than in the mismatch distribution ($\sigma^2_{\text{mismatch}} = 10 \times 0.2 - 10 \times 0.2^2 = 1.6$). Because the variance of the distribution does not scale linearly with its mean, parameter values of p do exist that produce greater mean with either higher, equal, or lower variability in match compared to mismatch distributions. Figure 3 gives the results of a simulation study that demonstrates the feasibility of this model in capturing all

Figure 3
Simulations With the Feature-Matching Similarity Model



Note. Gaussian similarity signal distributions generated by a feature-based matching model. The direction of the inequality of variances is determined by the feature-matching rate parameter p. The number of features in all cases is 10 ($n = 10$). Panel (a): Higher variability in match compared to mismatch distribution, (b) equal variability in match and mismatch distributions, and (c) higher variability in mismatch compared to match distribution. See text for more details.

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possible directions of the inequality of variances, using various values of p. Note that in all the simulated cases, the number of features is constant ($n = 10$). The left panel presents a case, where variability is higher in the match compared to mismatch distribution ($\sigma^2_{match} = 2.4$, $\sigma^2_{mismatch} = 1.6$). The middle panel presents a case of equal variability in the two distributions ($\sigma^2_{match} = 2.1$, $\sigma^2_{mismatch} = 2.1$). The left panel illustrates a case with higher variability in the mismatch compared to match distribution ($\sigma^2_{match} = 0.9$, $\sigma^2_{mismatch} = 2.5$).

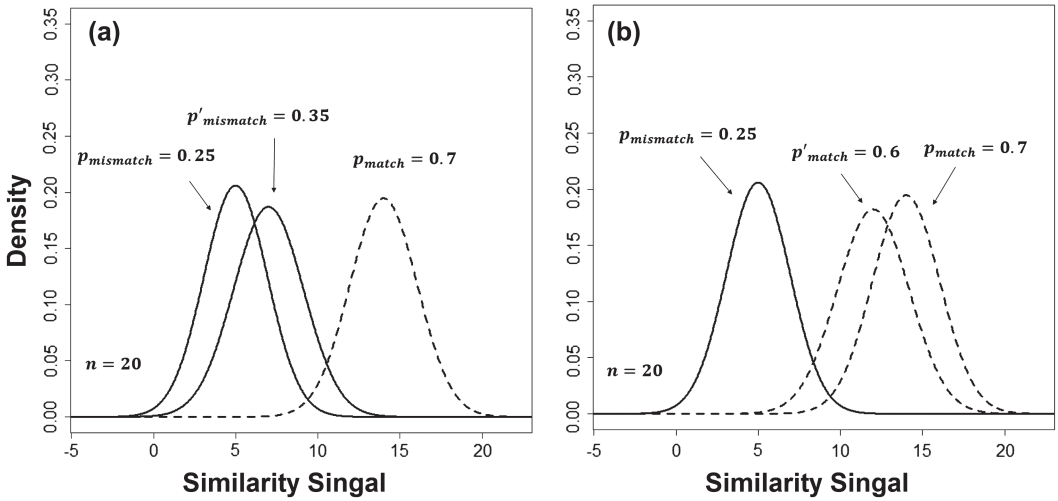
The proposed feature-matching model can explain various central findings in the literature. For example, the observation that match and mismatch conditions are affected differently by various experimental factors. Mismatch but not match accuracy is decreased when one face appears in the format of a passport pic (McCaffery & Burton, 2016) or when participants operate under time pressure (Bindemann et al., 2016). In the study by Wirth and Carbon (2017), manipulation of specific facial features hampered performance on mismatch trials by increasing false-alarm rates, but left the rates of correct matching (hits) intact. Selective influence in all of these cases can be understood in terms of specific alteration of the face-matching rate p. I run several simulations to demonstrate the capability of the feature-matching model to do exactly this. In one simulation, I generated the binomial and their approximated Gaussian distributions for match and mismatch conditions with feature-matching rates of $p_1 = 0.7$ and $p_2 = 0.25$, respectively. I then increased the feature-matching rate for mismatch trials p_2 by 0.1 ($p'_2 = 0.35$). The idea is that under these experimental conditions, more features are (incorrectly) matched. The results can be seen in Panel A of Figure 4. The variance of the new mismatch distribution increased from $\sigma^2 = 3.75$ to $\sigma^2 = 4.55$, while the variance of the match distribution remained the same ($\sigma^2 = 4.20$). As noted earlier, the location of the pertinent distribution is also affected by changes in p. In this case, the mismatch distribution is shifted a bit to the right (from $\mu = 5$ to $\mu = 7$). Assuming that the criterion remains invariant, such selective influence can be responsible for the observed decline in accuracy with mismatch trials (Wirth & Carbon, 2017).

In several other studies, match, but not mismatch, accuracy is hampered, for example, when the quality of one of the images is degraded (Strathie & McNeill, 2016) or when base rates of match stimuli are altered (Papesh et al., 2018). To simulate this type of selective influence, I, again, generated a pair of binomial distributions and their approximated Gaussian distributions, one for match and the other for mismatch conditions, with feature-matching rates of $p_1 = 0.7$ and $p_2 = 0.25$, respectively. I then decreased the feature-matching rate p_1 of the match condition by 0.1 ($p'_1 = 0.60$) under the assumption that under this type of experimental manipulations, feature-matching rate decreases for match stimuli. The results can be seen in Panel B of Figure 4. The variance of the new match distribution increased from $\sigma^2 = 4.20$ to $\sigma^2 = 4.80$, while the variance of the mismatch distribution remained the same ($\sigma^2 = 3.75$). The location of the match distribution shifted a bit to the left (from $\mu = 14$ to $\mu = 12$). Assuming that the criterion remains the same, this type of selective influence can generate the observed decline in the accuracy of match trials (Papesh et al., 2018; Strathie & McNeill, 2016).

Another central phenomenon that can be explained by the feature-matching model is the zero correlation observed between match and mismatch unfamiliar items (Megreya & Burton, 2007). To simulate this phenomenon, I assume that the number of facial features is held fixed across conditions ($n = 10$) and that there are 100 match and 100 mismatch items. To create variability in the face-matching rate p of these items, I have independently sampled values of binomial rates p_1 and p_2 for match and mismatch items from two independent beta distributions. The beta distribution is a family of continuous probability distributions defined on the interval [0,1]. Its shape is determined by two parameters α and β. Its mean is $\mu = \frac{\alpha}{\alpha+\beta}$, and its variance is $\sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)(\alpha+\beta+1)}$. I fixed $\alpha = 8$ and then varied β to achieve a given proportion of p. These simulations allowed me to create item variability by drawing item-level binomial p values from a β distribution with a relatively high mean for matching faces. In that case, some matching faces would have a higher probability of feature matching with the same identity than other face pairs. In

Figure 4

Simulations With the Feature-Matching Similarity Model



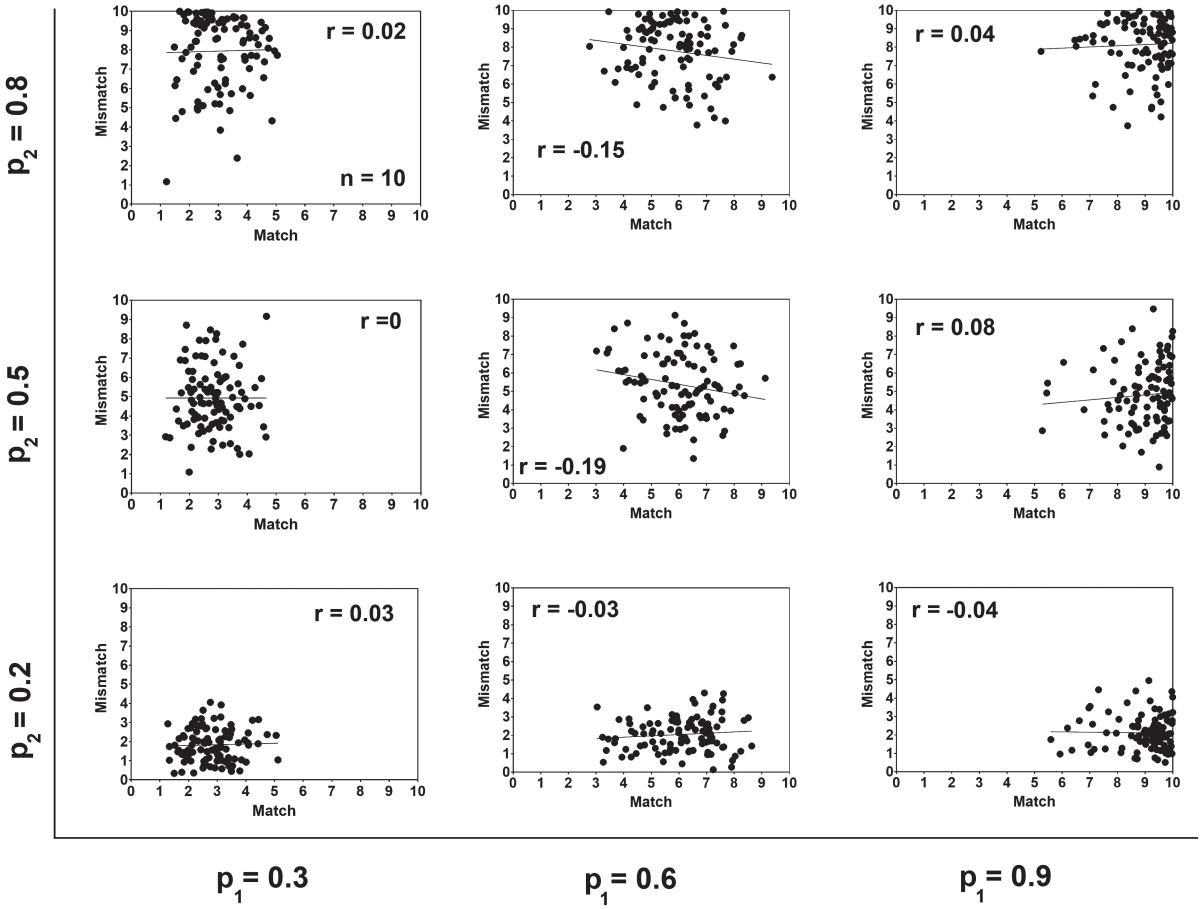
Note. Selective influence on either of the distributions is demonstrated. In all cases, the feature number is 20 ($n = 20$). Panel (a): Selective influence on the mismatch distribution. Panel (b): Selective influence on the match distribution.

addition, I could independently draw values from a beta distribution with a relatively low mean for nonmatching faces. In that case, some mismatching faces would have a higher probability of feature matching than other face pairs. The question of interest is whether the strengths of similarity signals produced by the binomial processes in the match and mismatch conditions would exhibit a significant correlation. To allow for broad sampling of the parameter space, I have created a 3×3 factorial design with $\bar{p}_1 = 0.3$, 0.6, and 0.9 and $\bar{p}_2 = 0.2$, 0.5, and 0.8. I then computed the correlation between the similarity signal produced by the binomial processes for match and mismatch items per each of the resulting nine combinations of parameter values in this factorial design. The outcomes can be seen in Figure 5. As can be noted, the correlations were mostly null, ranging between −0.19 and 0.08 and statistically insignificant, an outcome that is consistent with the empirical findings (Megreya & Burton, 2007).

It is important to note that the model assumes that each face image is represented as a vector in a featural face space, and similarity is computed between images and not between norm faces. This is in contrast to extant geometric models (Valentine, 1991) in which each face is represented by a single vector or point. In such models, within-identities variation is often not taken into consideration. Thus, the feature-matching model can accommodate violations of the distance axioms, such as the axiom of minimality (Tversky, 1977), when an image of Face A is judged as more similar to image of Face B than to another image of Face A. Most extant geometric models of face perception cannot account for the violation of the minimality axiom in face recognition (White et al., 2022), at least not without adopting ad hoc assumptions (Busey, 2005). This is mainly because they do not take into consideration within-identities variability. For example, in applications of the generalized context model (Nosofsky, 1988) to faces (Trujillo & Anderson, 2023), each face is represented as a single exemplar, and face typicality is computed as the sum of distances from all other faces. In the central prototype model (Valentine, 1991), each face is represented by a vector according to its distance from a norm face that is located at the origin of space. The upshot is that simultaneous modeling of both within-identities and between-identities variability requires a framework that inherently assumes stochastic representation of facial identity. Such a framework might be found, in addition to the present featural model, also in elaborations of SDT such as the general recognition theory (GRT; Ashby & Townsend, 1986). GRT is a multidimensional extension of SDT (Algom & Fitousi, 2016; Fitousi, 2018; Kadlec,

Figure 5

Simulations With the Feature-Matching Similarity Model



Note. Correlations between the strengths of similarity signals produced by match and mismatch items according to a 3×3 factorial design. The *x*-axis presents the values of the parameter p_1 in the match condition, while the *y*-axis presents the values of the parameter p_2 in the mismatch condition.

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1995; Kadlec & Townsend, 1992; Macmillan & Creelman, 2004; Soto et al., 2017). GRT has been extended by Ashby and Perrin (1988) to account for similarity relations among objects in a way that allows for violations of distance axioms, including minimality. The GRT can serve as a viable face space that generates the match and mismatch distributions postulated by the current UVSD model of face matching. However, a full elaboration of this theoretical venue will be pursued elsewhere.

Relations to Alternative Same–Different SDT Models

A comment is in order regarding the relations of the proposed single-system UVSD model to related same–different SDT models. Two primary same–different models are discussed in the literature (DeCarlo, 2013). According to the "independent observation" model (Rousseau, 2001), the observer makes separate covert decisions for each event as whether it belongs to A or B and then responds "same" for the AA and BB decisions and "different" for AB and BA decisions. Application of this model to face matching implies that the observer judges each face separately as belonging to a given identity (A or B) and then responds according to this model's decision rule. Although this is a valid theoretical model of face matching, in practice, it is not a viable one. The canonical face-matching task employs multiple identities (more than two events/faces), and therefore a practical application of this model requires estimation of dozens of distributions. In contrast, a more parsimonious approach—the "differencing" model (Macmillan & Creelman, 2004)—postulates that the observer bases their decision on differenced random variables $\psi_2 - \psi_1$. The proposed UVSD is a case in point because here the operation of differencing results in a continuous similarity signal that is elicited by each face pair and not by individual faces. As in a paired *t* test, this allows us to circumvent the problem of defining each face in the face space by relating to differences rather than nominal values. Notably, the differencing has also a representation in the original face space. It is possible to interpret the decision strategy without recourse to "differencing" by assuming that the observer is placing a "yardstick" (Lee et al., 2007) around the perceptual realization of the first (left-hand) face and then making a decision of "mismatch" according to the following decision rule:

$$Y = 1 \quad \text{if} \begin{cases} \psi_2 > \psi_1 + \tau \\ \text{or} \\ \psi_2 < \psi_1 - \tau \end{cases}, \tag{8}$$

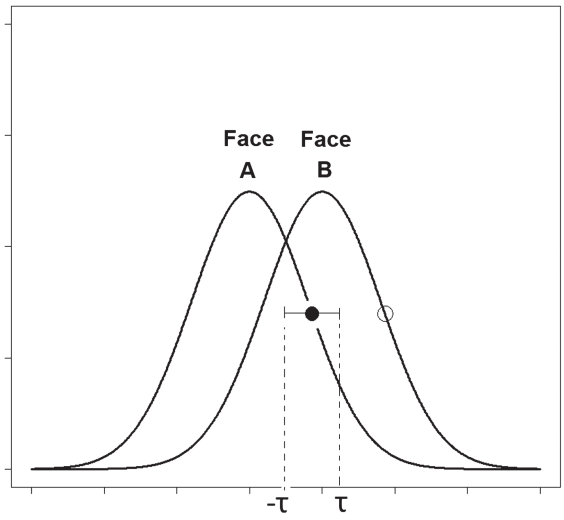
else $Y = 0$, where 0 is "match" and τ is the length of the yardstick, and 2τ is total length. If the perceptual effect of the second (right-hand) face falls within the yardstick, then the decision is "match," else it is "mismatch." An illustration of this model can be seen in Figure 6. It appears that the "differencing" and "yardstick" approaches yield mathematically equivalent models for the same–different task (see DeCarlo, 2013).

Relating Confidence and Accuracy in Face Matching

The UVSD model can provide valuable insights into the relations between confidence and accuracy (Peirce, 1877). Curiously enough, confidence ratings have been only sparsely used in face-matching studies (but see Smith et al., 2021). This is unfortunate given that confidence is a central determinant of judgment and decision making (Pleskac & Busemeyer, 2010) and has been recognized as such in related areas of eyewitness identification (Gronlund et al., 2014; Wixted & Mickes, 2014; Wixted & Wells, 2017). The current UVSD model provides a comprehensive theoretical account of the role of confidence in face matching. The model's predictions regarding confidence are subject to a set of rigorous empirical tests.

Figure 6

An Illustration of the Yardstick Decision Model of Face Matching



There are only few studies that did test for confidence in face matching. In one study, White et al. (2014) found that mean confidence ratings was higher for correct than incorrect decisions. In yet another study, Bruce et al. (1999) reported strong relations between accuracy and confidence in a one-to-many face-matching task. But these results cannot be accepted as fully diagnostic because the target in this study was always present. Moreover, not until the study by Stephens et al. (2017) has the *positive–negative asymmetry* been investigated. This asymmetry is documented in the domains of eyewitness identification (Brewer & Wells, 2006; Sporer et al., 1995; Weber & Brewer, 2004) and in standard list-learning studies of memory (Mickes et al., 2011). The asymmetry entails strong linear confidence–accuracy relationships for positive decisions ("This is the person I saw before") and weak or no relations for negative decisions ("This is not the person I saw before") (Brewer & Wells, 2006; Weber & Brewer, 2004). One possible explanation of this phenomenon is that in positive decisions, people can base their judgment on the similarity of the test item to the one in memory, whereas in negative decisions, such a comparison is not available. However, in face matching, where this asymmetry translates to a *match–mismatch asymmetry*, there is no such obstacle, and therefore, it is predicted that a negative–positive asymmetry should not be found. The study by Stephens et al. (2017) tested exactly this conjecture, using both ROC curves and confidence–accuracy plots. In their Experiment 1, the authors did find evidence for the positive–negative asymmetry, but in their Experiment 2, when match and mismatch face pairs were equated on difficulty, the asymmetry was absent. As can be noted, at this stage of research, the relations between confidence, accuracy, and similarity in face matching are not well understood. The available research suggests that these variables may be related to some degree, but the underlying mechanisms are unknown. The current effort offers

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a comprehensive account of the mechanisms that relate accuracy, confidence, and similarity.

Confidence–accuracy relationship reflects an important aspect of performance. They may not be easily deduced (Roediger & DeSoto, 2014), partly due to the existence of various types of measurements (Pleskac & Busemeyer, 2010); but once a satisfactory model is determined, the attendant measures can contribute invaluable insights into the psychological processes in question (Nguyen et al., 2017). A well-known descriptive formula has been proposed by Peirce and Jastrow (1884):

$$\overline{\text{con}\,f} = \beta \times \log\left[\frac{P(R_A|S_A)}{P(R_B|S_A)}\right], \tag{9}$$

where β is a scaling parameter and $P(R_A|S_A)$ and $P(R_B|S_A)$ are probabilities of responding correctly and incorrectly, respectively, to the appearance of stimulus of Class A. According to this formula, confidence is related to the ability of the observer to discriminate between two stimuli. In terms of SDT, this equation relates confidence to the ratios of hits and false alarms. Similar ideas have been imported to SDT frameworks (Pleskac & Busemeyer, 2010), helping researchers to link confidence ratings to SDT constructs. In traditional SDT approaches, confidence is directly related to response criteria (Macmillan & Creelman, 2004). It is often assumed that by placing a response criterion, the observer also assigns a confidence rating to his or her judgment (see Figure 2). In the current UVSD face-matching model, confidence is directly related to the observer's facility at discriminating between match and mismatch face pairs.

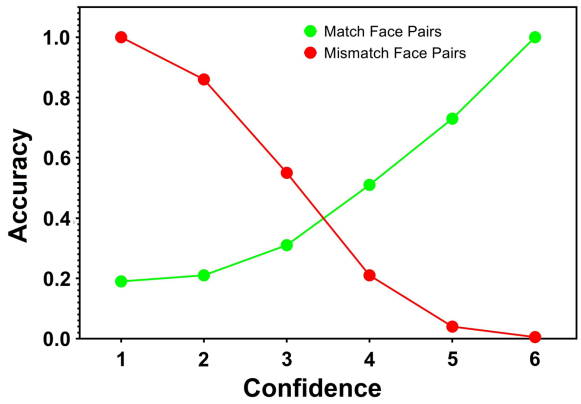
One way of modeling the predicted confidence–accuracy relations is by assuming that (a) there are separate accuracy–confidence functions for match and mismatch face pairs; (b) these functions are by-products of the interplay between the match and mismatch distributions and the location of the response criteria; (c) accuracy levels at a given confidence rating/response criteria C_i level are determined by the likelihood ratio of the accumulated evidence for correct match (mismatch) and incorrect match (mismatch) responses; (d) for match face pairs, evidence gets stronger as the face pair produces stronger identity signals, an outcome that is also associated with the placement of more conservative response criteria and higher confidence ratings. The evidence is quantified as the cumulative probability (i.e., area under the curve) from $-\infty$ to the corresponding response criteria c_i. For mismatch pairs, the evidence gets stronger as the face pair elicited weaker identity signals. The evidence is quantified as the cumulative probability from $+\infty$ to the corresponding response criteria c_i under the mismatch distribution; and lastly, (e) accuracy rates for high-confidence ratings (1 and 6) are set to 1 because accumulated evidence at these areas equals the integral of the probability distribution function from $+\infty$ to $-\infty$, which is practically 1. The likelihood ratios for confidence ratings 2–5 are associated with the likelihoods at response criteria C_1–C_5, respectively. These are quantified separately for match and mismatch distributions:

$$P_{\text{match}}(\text{correct})|(c_i) = \frac{\Phi_2(c_i)}{\Phi_1(c_i)}$$
$$P_{\text{mismatch}}(\text{correct})|(c_i) = \frac{1-\Phi_1(c_i)}{1-\Phi_2(c_i)}, \tag{10}$$

where $\Phi_2(c_i)$ and $\Phi_1(c_i)$ are the cumulative probabilities under the match and mismatch distributions, respectively, for a given c_i response criterion. The accuracy rates for the "high" confidence ratings (1 = *mismatch high confidence* and 6 = *match high confidence*) are 1, and they decline as confidence ratings (and corresponding response criteria) move toward "medium" and "low" ratings. This approach predicts that as response criteria become more conservative, the assigned confidence ratings should also increase because accumulated evidence for a correct match increases, while the evidence for an incorrect match decreases, and as a result, accuracy rates should improve (Roediger & DeSoto, 2014). The intuition for these predictions is that match face pairs that produce stronger identity signals are more likely to be call a "match" by the observer and are also more likely to be ascribed high-confidence judgment ratings. As for mismatch face pairs, the stronger is the identity signal elicited, the more likely that the observer will ascribe higher confidence for (an incorrect) "match" decision, which will result in lower accuracy rates. Thus, the predictions for correct responses are that accuracy will be high when confidence is high and will decline as confidence decreases, while for incorrect choices, accuracy will be low when confidence is high and will increase as confidence declines. This pattern entails a positive correlation between confidence and accuracy. Note, however, that because the ratings range from 1 to 6, with 1 = *mismatch high confidence* and 6 = *match high confidence*, the correlation for mismatch face pairs is plotted as a negative slop line. Figure 7 presents the simulated pattern with mismatch ($\mu = 0$, $SD = 1$), match ($\mu = 2$, $SD = 1.6$) distributions, and response criteria ($C_1 = -1$, $C_2 = 0$, $C_3 = 1$, $C_4 = 2$, and $C_5 = 3$). These are the exact parameters used in the simulation depicted in Figure 2. The resulting pattern of accuracy–confidence is that of two crossing lines, which meet at the point where confidence rating is approximately 3.5 and accuracy is about 0.4 correct.

Figure 7

UVSD Predictions: Accuracy Rates as a Function of Confidence Ratings in the Face-Matching Task



Note. The function is also called confidence–accuracy plot (see text). The participant is asked to rate his/her confidence in the decision according to the following: 1 = *mismatch high confidence*, 2 = *mismatch medium confidence*, 3 = *mismatch low confidence*, 4 = *match low confidence*, 5 = *match medium confidence*, and 6 = *match high confidence*. The pattern is derived from a simulation with the following distributional parameters: mismatch distribution ($\mu = 0$, $SD = 1$), match distribution ($\mu = 2$, $SD = 1.6$), and response criteria ($C_1 = -1$, $C_2 = 0$, $C_3 = 1$, $C_4 = 2$, and $C_5 = 3$). For more details, see text. UVSD = unequal variance signal detection. See the online article for the color version of this figure.

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The current framework is consistent with the study of confidence–accuracy relations in the face recognition and eyewitness memory literature (Brewer & Wells, 2006; Weber & Brewer, 2004; Weber et al., 2013). This approach suggests that there is some level of correspondence between confidence and the proportion of correct decisions. In this sense, confidence can, to some extent, predict accuracy level. A useful tool for assessing these relations has been the confidence–accuracy plot. This tool is applied to the aggregate results (across observers), by plotting the mean proportion correct as a function of the mean reported confidence. This plot can answer the question of how accurate is someone given that her confidence is *x*. The confidence–accuracy relations derived from the UVSD and presented in Figure 7 are basically confidence–accuracy plots for match and mismatch trials.

Relating Accuracy and Similarity in Face Matching

Similarity plays a crucial theoretical role in many canonical models of face recognition (Nosofsky, 1988; Trujillo & Anderson, 2023; Valentine, 1991; Vanpaemel & Storms, 2008). In the proposed UVSD model, a continuous perceptual similarity signal is associated with each face pair and, along with a criterion placement operation, dictates the rates of hit and false alarm in the matching task. Heretofore, the variable of similarity in the UVSD has been assumed to act as the driving force behind face-matching decisions, but no means for testing its presence and its influence on accuracy and confidence has been offered yet. One way of precisely assessing the role of similarity in face-matching accuracy and confidence is asking an independent group of observers to rate their perceived similarity/dissimilarity of the face pairs (Fysh & Bindemann, 2023; Rice et al., 2013)[1] by choosing from the following labels: 1 = *high dissimilarity*, 2 = *medium dissimilarity*, 3 = *low dissimilarity*, 4 = *low similarity*, 5 = *medium similarity*, and 6 = *high similarity*. The question of interest is whether these similarity ratings bear consistent relations with the accuracy and confidence ratings that are collected in the actual matching task. The UVSD model predicts that they do and allows for qualitative and quantitative hypotheses testing. Two hypotheses follow from the proposed UVSD model: (a) Similarity ratings are expected to be positively correlated with confidence ratings and (b) similarity ratings predict accuracy rates in match and mismatch trials. The logic is as follows:

I assume that, just as in the confidence-rating task, participants accomplish the similarity-rating task by placing $N - 1$ decision bounds $s_1, s_2 \ldots s_{N-1}$, such that $s_1 \leq s_2 \leq \ldots s_{N-1}$. In the present case, six degrees of similarity yield five similarity response criteria. Following this scheme, an identity signal that falls below s_1 produces the response "high dissimilarity," identity signal that falls above s_1 and below s_2 results in the response "medium dissimilarity," and so forth. I also make the assumption that response criteria in the similarity task ($s_1 \leq s_2 \leq \ldots s_{N-1}$) and response criteria in the confidence task ($c_1 \leq c_2 \leq \ldots c_{N-1}$) are not necessarily identical, but do exhibit a high level of correspondence. Such a correspondence is a plausible assumption if the psychological face space remains invariant across tasks, and if a comparable similarity signal is evoked by the observer irrespective of the task at hand. In that case, response criteria preserve their ordering relations with respect to the decision variable. Thus, shifting the response criteria of the confidence task either to the left or to right should not alter their degree of correlation with the similarity response criteria, just as increasing or decreasing a fixed amount from one variable does not alter its correlation with another. In addition, the model is agnostic with respect to the exact causal mechanism by which similarity and confidence are related. It is, however, more likely that similarity derives confidence (Shepard, 1987) due to the precedence of perceptual processes over higher cognitive processes. This entails that as face pairs become more similar/dissimilar, observer's confidence in their matching decision increases. All of this entails that mean experienced similarity ratings are expected to be positively correlated with their corresponding mean confidence ratings.

The second hypothesis concerns the relations between accuracy in the matching task and mean similarity ratings. Under the assumption that confidence and similarity response criteria are positively correlated, similarity ratings and accuracy should exhibit a pattern that is comparable to that expected for confidence and accuracy. The predicted pattern (see Figure 8) consists of a monotonically increasing function for match face pairs, with increasing accuracy rates as a function of similarity, and a monotonically decreasing function for mismatch face pairs, with decreasing accuracy rates as a function of similarity. Accordingly, match face pairs that are judged as highly similar and rated with high confidence should lead to high hit rates, whereas mismatch face pairs that are judged as highly similar and rated with high confidence should produce high false-alarm rates. The unified confidence–similarity UVSD model offers a single-system mechanistic explanation of these relations.

As shown in the previous section, accuracy in the face confidence-rating matching task depends on the accumulated evidence for correct decision to a given response criterion, which is computed separately under the match and mismatch distributions (see Equation 10). Commensurate with this logic, highly similar or highly dissimilar face pairs (ratings of 1 or 6) are likely associated with perfect accuracy, which is expected to decline as similarity/dissimilarity decreases. In addition, separate accuracy–similarity functions exist for match and mismatch face pairs. Accuracy rates for similarity ratings 2–5 are associated with their corresponding accuracy levels, which are produced by similarity response criteria s_1-s_5, respectively. These are quantified separately for match and mismatch distributions according to Equation 10. Thus, for match face pairs, accuracy gets higher as similarity ratings increase, whereas for mismatch pairs, accuracy decreases as similarity ratings increase. It should be noted that the relations between similarity and confidence are correlative likely due to a high degree of correspondence between the response criteria in the two tasks.

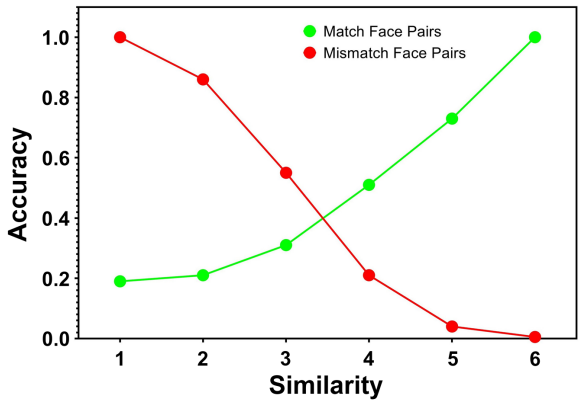
Alternative Models of the Face-Matching Task

An often underplayed but fundamentally critical issue in face matching concerns the nature of the internal representations that guide performance. SDT models assume continuous representations, an assumption that is reflected in curvilinear ROCs (Green & Swets, 1966). An equally valid theoretical assumption is that face matching is underlined by *all-or-none* perception (Broadbent, 1966; Cook & Wixted, 1997; Nachmias & Steinman, 1963; Province & Rouder, 2012; Qin et al., 2001; Yonelinas, 1994). Namely, the observer either

[1] Note that objective measures of similarity, such mutual information between the two images, are inappropriate because they are fixed, and as such cannot account for dynamic effects of learning and familiarization.

Figure 8

UVSD Predictions: Accuracy Rates in the Face-Matching Task as a Function of Similarity Ratings



Note. The participant is asked to rate similarity according to the following: 1 = *high dissimilarity*, 2 = *medium dissimilarity*, 3 = *low dissimilarity*, 4 = *low similarity*, 5 = *medium similarity*, and 6 = *high similarity*. The pattern is derived from a simulation with the following distributional parameters: mismatch distribution ($\mu = 0$, $SD = 1$), match distribution ($\mu = 2$, $SD = 1.6$), and similarity response criteria ($S_1 = -1$, $S_2 = 0$, $S_3 = 1$, $S_4 = 2$, and $S_5 = 3$). For more details, see text. UVSD = unequal variance signal detection. See the online article for the color version of this figure.

perceives an identity-match signal or not, and he or she bases their decision on this binary information.[2] An entire class of models adopts the all-or-none assumption (Blackwell, 1963; Swets et al., 1961). These models predict linear rather than curvilinear ROC curves (Green & Swets, 1966; Macmillan & Creelman, 2004; Yonelinas & Parks, 2007). It should be noted though that threshold models can produce various nonlinear ROCs by adding parameters or thresholds (Krantz, 1969; Malmberg, 2002). However, the latter approach is rare because it requires additional parameters (Hilford et al., 2002).

It is not only important to demonstrate that the UVSD is a viable model of performance, but also to show that it outperforms other alternative models. The simplest model of the face-matching task is the binominal model. This is not a psychological model but only a statistical benchmark against which other models can be tested. The model consists of two free parameters, which are the hit and false-alarm rates. The model is given as:

$$\begin{aligned} Y_h &\sim B(p_h, N_s) \\ Y_f &\sim B(p_f, N_n), \end{aligned} \tag{11}$$

where Y_h and Y_f are the number of hits and false alarms, respectively; p_h and p_f are the true probabilities of hits and false alarms, respectively; and N_s and N_n are the total number of trials of face matches and face mismatches, respectively. The complementary probabilities of miss and correct rejection can be computed from the hit and false-alarm rates. The model can be analyzed by treating the components independently. For the two-choice paradigm (Experiments 1a and 2a), the model has two free parameters (p_h and p_f), while for the confidence-rating experiment data (Experiments 1b and 2b), the model has 10 parameters (p_h and p_f for each of the five confidence intervals).

The *equal variance signal detection* model (Macmillan & Creelman, 2004) is a closely related alternative to the UVSD. It has two parameters d and c, and the variances of the two normal distributions are equal and set to 1:

$$\begin{aligned} p_h &= 1 - \Phi(c - d') \\ p_f &= 1 - \Phi(c). \end{aligned} \tag{12}$$

Notably, this model cannot account for the match–mismatch dissociation. The match and mismatch distributions in this model are affected by two parameters, one of them is common to both. This suggests that the hit and false alarms are correlated. In the confidence-rating task, the equal variance SDT model has six parameters (d, c_1, c_2, c_3, c_4, and c_5).

The *high-threshold model* (Blackwell, 1963; Swets et al., 1961) assumes that perception is all-or-none (Province & Rouder, 2012). Thus, on a "signal" trial (face match), the observer can be in one of two mental states. He or she may either have detected face match, with probability d, or failed to do so, with probability $1 - d$. When observers fail to detect the face match, they still may guess that the faces match with probability g. In this model, hit can be generated by two events: (a) successful match detection and (b) detection of match fails yet the observer still guesses that the faces match. This possibility of hit occurs with probability $(1 - d) \times g$. Combining these two rates gives the probability of a hit: $p_h = d + (1 - d) \times g$. For mismatch trials, there is no match signal to detect. False alarms are generated only by guessing: $p_f = g$. The high-threshold model has two parameters, d and g. The first serves as a sensitivity measure, whereas the second serves as a bias parameter (see Figure 9). The model can be summarized as follows:

$$\begin{aligned} p_h &= d + (1 - d) \times g \\ p_f &= g. \end{aligned} \tag{13}$$

For the confidence-rating paradigm, the model has six parameters: a common sensitivity parameter and five bias parameters (d, g_1, g_2, g_3, g_4, g_5). The high-threshold model predicts straight-line isosensitivity plots with a slope of $1 - d$ and intercept d.

The *double high-threshold model* is a closely related alternative to the high-threshold model that relies on the notion of all-or-none mental states (see Figure 9). The model assumes that observers may enter a mismatch detection state in which they believe that the faces presented do not belong to the same person. For face match present trials, either match is detected with probability d or not. If match signal is not detected, observers guess. On no-match trials, observers either detect that no match exists with probability d or enter a guessing state. The probability of a hit is $d + (1 - d) \times g$, and the probability of a false alarm is $(1 - d) \times g$. This model can be summarized as follows:

$$\begin{aligned} p_h &= d + (1 - d) \times g \\ p_f &= (1 - d) \times g. \end{aligned} \tag{14}$$

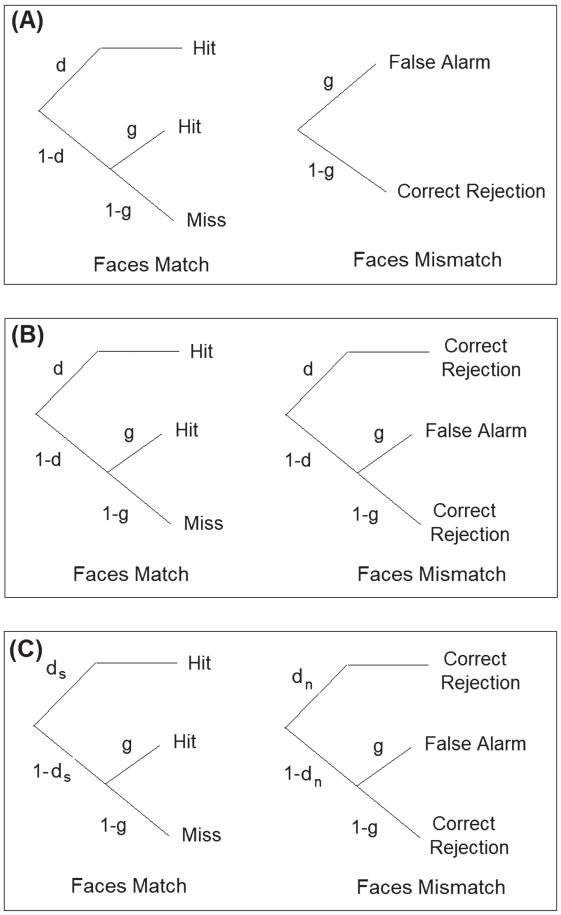
For the choice paradigm, the model has two parameters (d and g), whereas for the confidence-rating paradigm, this model has six parameters, a common sensitivity parameter plus five bias/guessing parameters for each confidence interval (d, g_1, g_2, g_3, g_4, g_5). The double high-threshold model predicts straight lines ROC with a slope of 1 and intercept d.

[2] The distinction between a continuous identity-signal and an all-or-none identity-signal is comparable to the distinction between familiarity and recollection made in the memory literature (Yonelinas, 1994).

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Figure 9

Three High-Threshold Models: (A) the High-Threshold Model, (B) the Double High-Threshold Model, and (C) the General High-Threshold Models



Next, consider the *general high-threshold model*. This model is a generalization of both the high-threshold and the double high-threshold models (see Figure 9). The three parameters are the sensitivity to signal (faces match) d_s, the sensitivity to noise d_n (faces mismatch), and the guessing bias g. The probability of a hit amounts to $d_s + (1 - d_s) \times g$, and the probability of a false alarm is $(1 - d_n) \times g$. Thus, this model has three free parameters in the simple choice paradigm, and it can be summarized as follows:

$$\begin{aligned} p_h &= d_s + (1 - d_s) \times g \\ p_f &= (1 - d_n) \times g. \end{aligned} \tag{15}$$

In the confidence-rating task, this model has seven parameters: two common sensitivity parameters plus five bias parameters (d_s, d_n, g_1, g_2, g_3, g_4, g_5). The general high-threshold model predicts straight isosensitivity lines with an intercept of d_s and a slope equal to $(1 - d_s)/(1 - d_n)$ (see Figure 10).

Last, consider the *low-threshold model* (Luce, 1963). According to this model, when a face match is presented, the observer either detects the match or its absence. Unlike the high-threshold models, there is no guessing in this model. The probability that the observer detects the match (signal) is d_s for match and d_n for mismatch (noise) trials. These parameters are sensitivity parameters. A response bias parameter b is added to these parameters to arrive at the hit and false-alarm rates. In the model, the bias is expressed as relative fraction rather than an absolute amount:

$$\begin{aligned} p_h &= \begin{cases} d_s + (1 - d_s) \times b & 0 \le b \le 1 \\ d_s + d_s \times b & -1 \le b \le 0 \end{cases} \\ p_f &= \begin{cases} d_n + (1 - d_n) \times b & 0 \le b \le 1 \\ d_n + d_n \times b & -1 \le b \le 0 \end{cases}. \end{aligned} \tag{16}$$

Thus, in a choice paradigm, this model has three parameters (d_s, d_n, and b), while in the confidence-rating task, this model is characterized by seven parameters: two common sensitivity parameters plus five bias parameters, one for each confidence interval (d_s, d_n, b_1, b_2, b_3, b_4, and b_5). The isosensitivity curves of this model are composed of two straight lines: one from the origin to point (d_n, d_s) and the other from point (d_n, d_s) to the point (1, 1). The former line (lower limb) results from $b < 0$, while the latter line (upper limb) results from $b > 0$ (see Panel c in Figure 10).

The Experiments

The unified confidence–similarity UVSD model of face matching is tested against its six alternative competitors. Predictions address the models' degree of fit to the data, as well as their ability to reproduce the empirical ROC curves (linear or curvilinear). Additional predictions concern the relations between confidence and accuracy, which are assessed by confidence–accuracy plots, and between similarity and accuracy. The predictions are tested in three experiments. Experiments 1a and 1b make use of the standard GFMT (Burton et al., 2010). Experiments 2a and 2b are applied to yet another standard test of face matching, the KFMT (Fysh & Bindemann, 2018). Experiment 3 is designed to test hypotheses regarding the relations between similarity ratings, confidence ratings, and accuracy rates with the GFMT. Data analyses are held with hierarchically nested modeling procedures (Ashby & Lee, 1991; Fitousi, 2014, 2021a), ROC analyses (Yonelinas & Parks, 2007), and confidence–accuracy plots (Weber & Brewer, 2004).

Experiment 1

The unified confidence–similarity UVSD model of face matching is assessed using one of the most canonical face-matching tests—the GFMT (Burton et al., 2010). In Experiment 1a, participants are tested in the original version of the GFMT. In Experiment 1b, a modified version of the GFMT is used, in which participants are asked to rate their confidence in their matching decisions. Central hypotheses of the unified confidence–similarity UVSD model are tested with respect to accuracy and confidence, using ROC curves and confidence–accuracy plots. The UVSD is compared to alternative models using hierarchically nested model fitting.

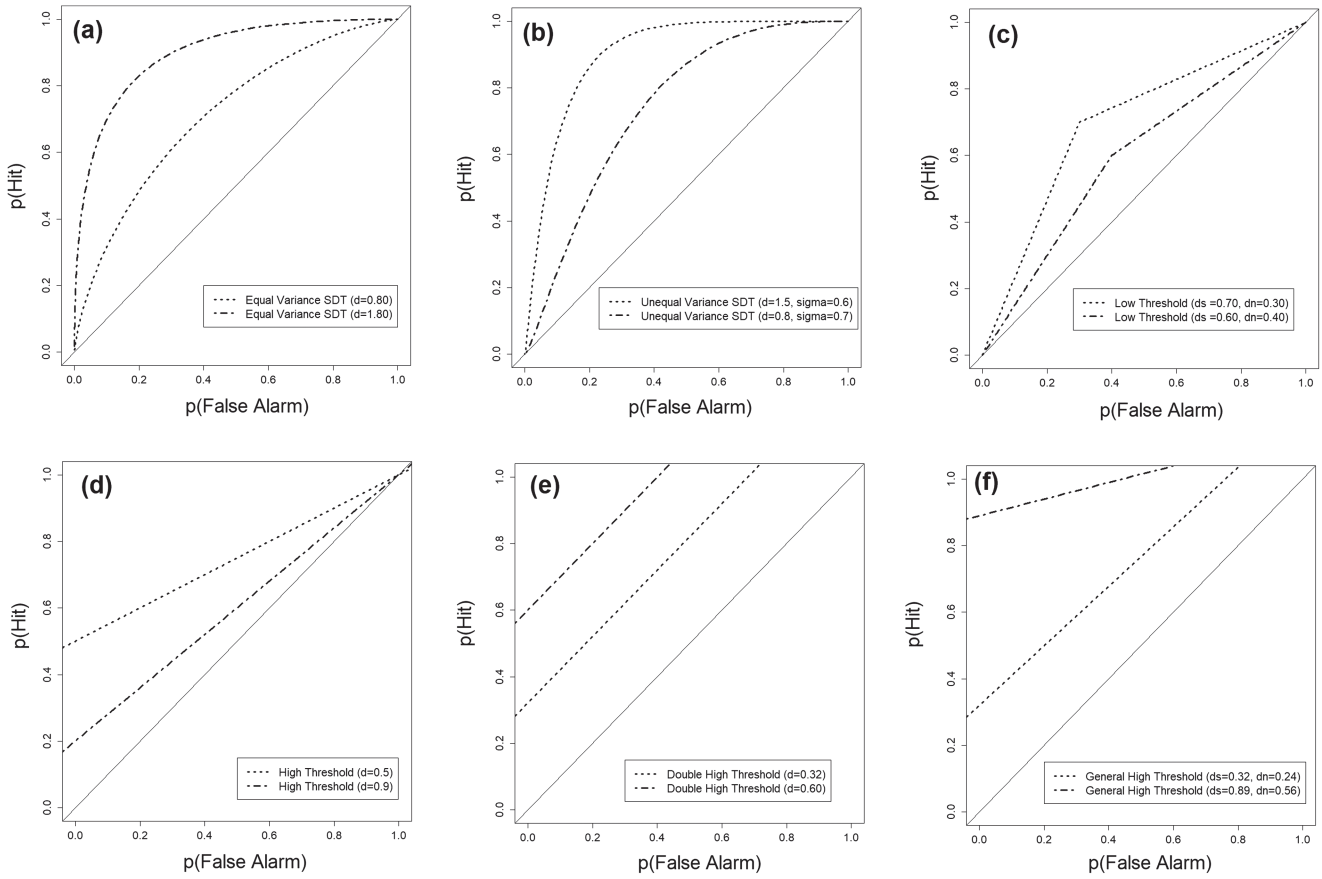
Method

Participants

A total of 96 participants participated in this study. There were 40 participants ($F = 24$, $M = 16$, $M_{age} = 25.08$, $SD = 3.70$) in Experiment 1a and 56 participants ($F = 46$, $M = 10$, $M_{age} = 24.3$,

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Figure 10
ROC Curves for Different SDT Models



Note. Simulated ROC curves for (a) equal variance signal detection model, (b) unequal variance signal detection model, (c) low-threshold model, (d) high-threshold model, (e) double high-threshold model, and (f) general high-threshold model. The values of parameters that were used in the simulations appear within each plot. ROC = receiver operating characteristics; SDT = signal detection theory.

SD = 5.11) in Experiment 1b. They were recruited from Ariel University pool of participants. Participants were compensated with a course credit.

All experiments were approved by the Ethical Committee of Ariel University (AU-SOC-DF-202208025). All participants gave their informed consent before participating in the experiments.

Stimuli and Apparatus

The set of stimuli in Experiments 1a and 1b consisted of 168 pairs of Caucasian unfamiliar faces. These faces comprise the long version of the GFMT (Burton et al., 2010). GFMT is a standard test for unfamiliar face matching. The faces in this test were photographed in the full-face view but with different cameras. The test is free for scientific purposes. Its normative data are well known (Burton et al., 2010). An example of a face pair from the GFMT is illustrated in Figure 11. The face pair stimuli consisted of high-quality gray scale digit photograph measuring approximately 12 × 7 cm. The faces conveyed neutral expression and pose. Half of the face pairs (84) comprised of identity-match pairs, while the other half were identity-mismatch pairs. The items in the GFMT are constructed such that the same identity (out of two) that shows up in a match pair also appears in a mismatch pair. This is important because it allows one to assess the relations between match and mismatch trials at the item level.

Procedure and Design

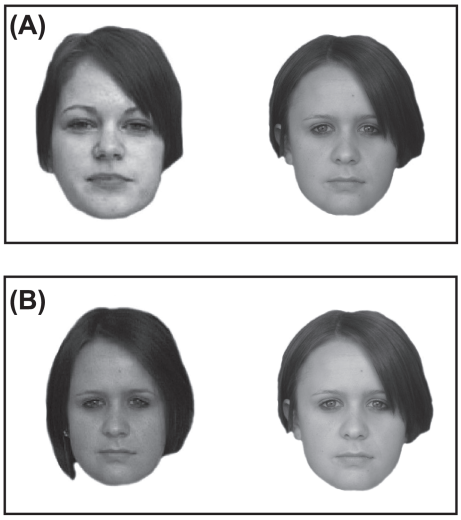
Experiments 1a and 1b were conducted online using *Qualtrics* software. Stimuli were presented randomly, one at a time. Experiment 1a employed a binary-choice task. Thus, in this experiment, the following question was presented under each face pair: "Do the two faces depict the same person or belong to different people?" Participants responded "same" or "different" by clicking on their choice. Experiment 1b administrated a confidence-rating task. In this experiment, the following question was presented: "Do the two faces depict the same person or belong to different people? Please indicate your confidence in your answer." Participants responded by clicking one of six possible choices. The choices were placed along a horizontal line from left to right starting with 1 = *different high confidence*, 2 = *different medium confidence*, 3 = *different low confidence*, 4 = *same low confidence*, 5 = *same medium confidence*, and 6 = *same high confidence*.

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Figure 11

Experiments 1a and 1b: Examples of Face Pairs From the GFMT (Burton et al., 2010)



Note. Panel A: The faces mismatch. Panel B: The faces match.

Results

Experiment 1a

In this experiment, participants gave binary responses. Overall accuracy level in the experiment, as measured by probability correct $p(c)$, amounted to 0.89 ($SD = 0.30$), which is quite high for a face-matching task, but is in agreement with the published results of the GFMT (Burton et al., 2010). Accuracy levels in match trials (0.87, $SD = 0.0056$) were slightly lower than those in mismatch trials (0.91, $SD = 0.0048$), $t(39) = -1.82$, $p = .03$. The canonical measures of hit and false alarm amounted to 0.87 and 0.084, respectively. Next, recount that previous studies have not found any relations between match and mismatch performance (Megreya & Burton, 2007) and that this pattern has been often raised to support the claim that match and mismatch are governed by independent processes (Fysh & Bindemann, 2023). To test whether this pattern is replicated in the current experiment, I computed the correlation between performance in match and mismatch trials at the *item level*. The Pearson correlation coefficient was close to zero, $C_p = 0.005$, $t(82) = 0.05$, $p = .96$. At the observer level, the Pearson correlation coefficient was also null, $C_p = 0.06$, $t(38) = 0.4$, $p = .67$. These results are in agreement with previous studies (Megreya & Burton, 2007; Megreya et al., 2011).

Experiment 1b

In this experiment, participants were asked to rate their confidence. Overall accuracy level in the experiment, as measured by probability correct $p(c)$, amounted to 0.86 ($SD = 0.34$), which is quite similar to that observed in Experiment 1a. Accuracy levels in match trials (0.91, $SD = 0.0041$) were not higher than those in mismatch trials (0.81, $SD = 0.0056$; $t < 1$). The canonical measures of hit and false-alarm rates amounted to 0.91 and 0.18, respectively. No relations, whatsoever, between match and mismatch performance were found in the data. The Pearson correlation between accuracy in match and mismatch trials at the *item level* was close to zero, $C_p = -0.09$, $t(82) = 0.05$, $p = .39$. At the observer level, the Pearson correlation coefficient was also null, $C_p = 0.08$, $t(54) = 0.4$, $p = .53$. These results are comparable to those observed in Experiment 1a, and they support the conclusion that switching from a binary-matching task to a confidence-rating task did not alter the main characteristics of the data.

Confidence and Accuracy

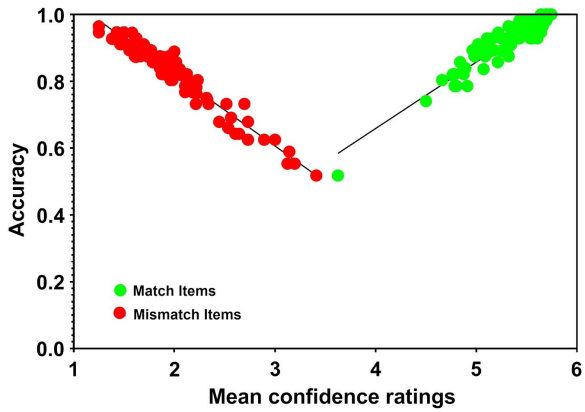
Mean confidence ratings and mean accuracy rates were computed for each face pair across all observers. Figure 12 presents the resulting confidence–accuracy plots for match and mismatch trials. As can be noted, the accuracy rates for both match, $R^2 = 0.88$, $t(82) = 25.40$, $p < .001$, and mismatch, $R^2 = 0.94$, $t(82) = 36.81$, $p < .001$, face pairs were linearly and positively correlated with confidence ratings. The pattern documented is consistent with the pattern predicted by the unified confidence–similarity UVSD model. These results show that face matching does not exhibit the positive–negative asymmetry of accuracy–confidence (Brewer & Wells, 2006; Sporer et al., 1995; Weber & Brewer, 2004). Instead, the current results adduce further evidence for symmetric confidence–accuracy relations (Stephens et al., 2017). These results support the notion that match and mismatch trials are processed by a single mechanism, best described by the unified confidence–similarity UVSD model.

Fitting SDT Models

In this section, I fit the UVSD model along with its six competitors to the empirical data. The data set consists of a matrix with the aggregate (across all participants) frequencies of trials in each condition (hit, miss, false alarm, and correct rejection), at each of five levels of confidence intervals. Models are fit to data using a *maximum likelihood* approach. The natural logarithm of the likelihood (*log likelihood*) is derived separately for each model. A maximum log-likelihood (ML) estimate is searched using an algorithm that maximized the

Figure 12

Experiment 1b: Confidence–Accuracy Plots Showing Mean Accuracy as a Function of Mean Confidence Ratings at the Item Level for Match and Mismatch Trials



Note. Confidence ratings 1–6 are as follows: 1 = *mismatch high confidence*, 2 = *mismatch medium confidence*, 3 = *mismatch low confidence*, 4 = *match low confidence*, 5 = *match medium confidence*, and 6 = *match high confidence*. See the online article for the color version of this figure.

log-likelihood function for the given data. To increase the accuracy of optimization, I use a *nested optimization* approach. Optimization of parameters specific to conditions (e.g., c_1, c_2 … c_n) is nested within parameters common across all conditions (e.g., d). This approach leads to far more accurate estimation. The seven models are represented as a tree (see Figure 13). At the top level is the binomial model with 10 parameters. The binomial model is the most general, and all the other (restricted) models are nested within it. At the next level are the general high-threshold, the low-threshold, and the free (unequal) variance signal detection models, each model with seven free parameters. At lower level, one can find the double high-threshold, the high-threshold, and the equal variance signal detection models, each model with six parameters. The lines connecting the models at different levels depict nesting relationships. So, for example, the equal variance signal detection model is nested within the free variance signal detection model (but not within the low-threshold model). Model selection (Myung & Pitt, 1997) was held with the Akaike information criterion (AIC; Akaike, 1973), which exacts penalty on additional free parameters:

$$\mathrm{AIC} = -2\log L(\theta*) + 2M, \tag{17}$$

where L is the likelihood of the model, θ^* are the maximum likelihood estimations of the parameters, and M is the number of free parameters. A related issue in model selection is that general models can show better absolute fit than do restricted models. This can occur either because the general model has more free parameters or because of its greater capacity to capture real aspects of the data (Ashby & Lee, 1991). To decide between these possibilities, one can use the likelihood-ratio test with the G^2 statistics (Ashby & Lee, 1991; Fitousi, 2014; Thomas, 2001). Consider the likelihood functions for a general and restricted model, L_g and L_r, with parameters p_g and p_r, respectively. Then:

$$G^2 = -2\log\left(\frac{L_r(\hat{p}_r; y)}{L_g(\hat{p}_g; y)}\right). \tag{18}$$

The statistic G^2 can be written more conveniently in terms of log likelihoods:

$$G^2 = -2(l_r[\hat{p}_r; y) - l_g(\hat{p}_g; y)]. \tag{19}$$

The G^2 statistics is distributed as a χ^2 with degrees of freedom equal to the difference in number of free parameters in the general and restricted models. If the G^2 is not significant, then we fail to reject the restricted model.

Table 1 summarizes the results of the model-fitting procedure. The binomial model (−log likelihood [−LL] = 18,868) exhibited the highest likelihood value. This was expected, given that it has the largest number of free parameters. The binomial model was hard to beat also in terms of AIC values and G^2. But it is important to note that the binomial is not a psychological model, and it served only as a reference for statistical testing with other models. Therefore, it is most appropriate to consider the next best-fitting models. These were the UVSD model (−LL = 18,881) and the equal variance SDT (−LL = 18,893). Next came the general high-threshold model (−LL = 19,103) and the double high-threshold model (−LL = 19,105), which produced relatively low fit. The high-threshold model (−LL = 19,885) provided the worst fit to the data, despite having a smaller number of free parameters.

Table 1
Experiment 1b: Summary of Fitting Results

Model	Parameters	−LL	AIC
Binomial	10	18,868	37,756
General high threshold	7	19,103	38,220
Low threshold	7	18,936	37,886
Unequal variance SDT	7	18,881	37,776
High threshold	6	19,885	39,782
Double high threshold	6	19,105	38,222
Equal variance SDT	6	18,893	37,774

Note. −LL = −log likelihood; AIC = Akaike information criterion; SDT = signal detection theory.

Comparisons of AIC values also favored the UVSD (AIC = 37,776) and the equal variance SDT (AIC = 37,774) models over the other models (excepting the binomial model that is not a psychological model). Model comparisons between general and nested models showed that one could not reject any of the general models in favor of the restricted models. This was indicated by the finding that all G^2s were significant (see Figure 13). If we jettison for a moment the binomial model from the competition due to the fact that it is not a genuine psychological model, then the results clearly suggest that the UVSD outperformed all the other relevant models. The UVSD provided the highest log likelihood and was preferred over restricted models in the hierarchy. These results support the UVSD and strengthen the conjecture that the internal representations that underlay face matching are continuous rather than discrete.

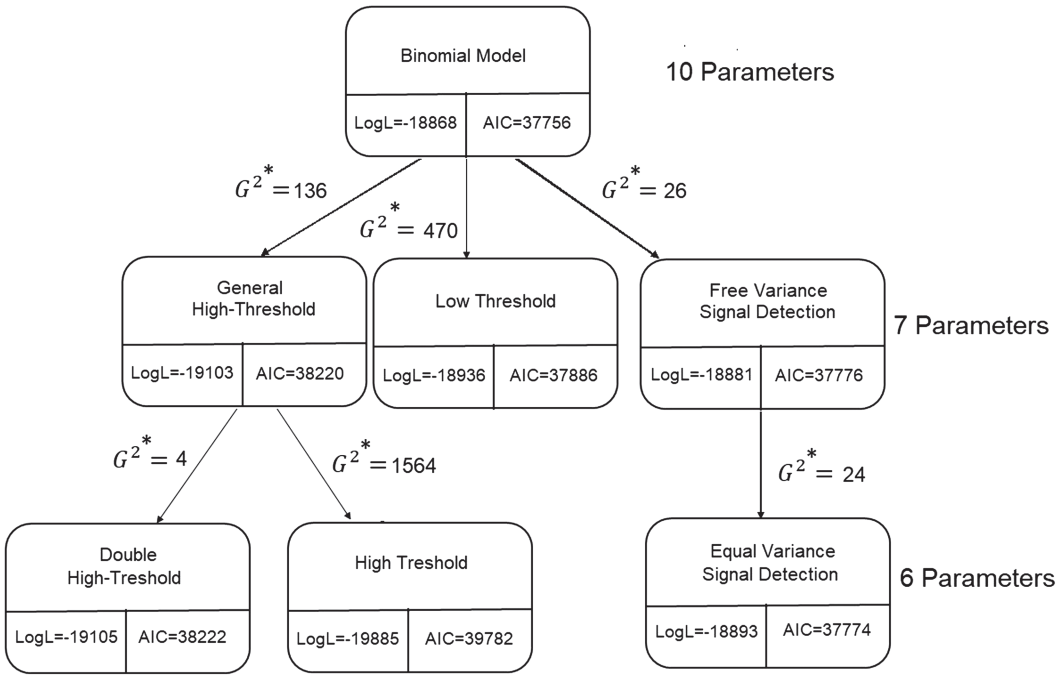
ROC Curves

To decide which of the models provide the best-fitting ROC curve to the empirical data, I first extracted the best-fitting parameters for each model (see Table 2). Notably, the UVSD parameters entail that the standard deviation of the match distribution ($\sigma = 0.85$) was smaller than that in the mismatch distribution. This suggests that in the GFMT set of stimuli, between-identities variance is higher than within-identities variance. As I noted earlier, the relations between the variances of the match and mismatch distributions are dictated by the choice of stimuli made by the experimenter. Natural image statistics computed under well-controlled conditions suggests that within-identities variability is higher than between-identities variability (Adjabi et al., 2020; Moses et al., 1994).

I have used the best-fitting parameters to generate the best-fitting ROC curve for each model. Figure 14 plots these modeled lines along with the empirical ROC points. The degree to which the predicted ROC lines fit the empirical data points can be visually examined. As can be noted, in Figure 14, the empirical ROC is curvilinear and, as such, refutes an entire class of models, predicting linear isosensitivity curves. The top panel of Figure 14 presents the ROC curves for the high-threshold, general high-threshold, and double high-threshold models. As can be noted, the high-threshold model gives the worst fit to the ROC points. This is expected, given its worst low-likelihood value. The double high-threshold and the general high-threshold models yield two quite similar isosensitivity lines, none of which is capable of accommodating the curvilinear shape of the empirical ROC. Next, consider the bottom panel of

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Figure 13
Experiment 1b: A Hierarchy of Seven Models



Note. Included are model comparison statistics G^2, LL, and AIC. See text for technical details. LL = log likelihood; AIC = Akaike information criterion.
* $p < .05$.

Figure 14, which illustrates the ROC cruves produced by the UVSD model, the equal SDT model, and the low-threshold model. As can be seen, both the UVSD and equal variance models are very adept at fitting the empirical ROC, as they follow the curvilinear shape of the ROC very closely. The low-threshold model fit is better than the high-threshold models, but not as good as the UVSD and the equal variance SDT models. It cannot perfectly accommodate the curvilinear shape of the empirical ROC. The ROC results clearly suggest that the UVSD

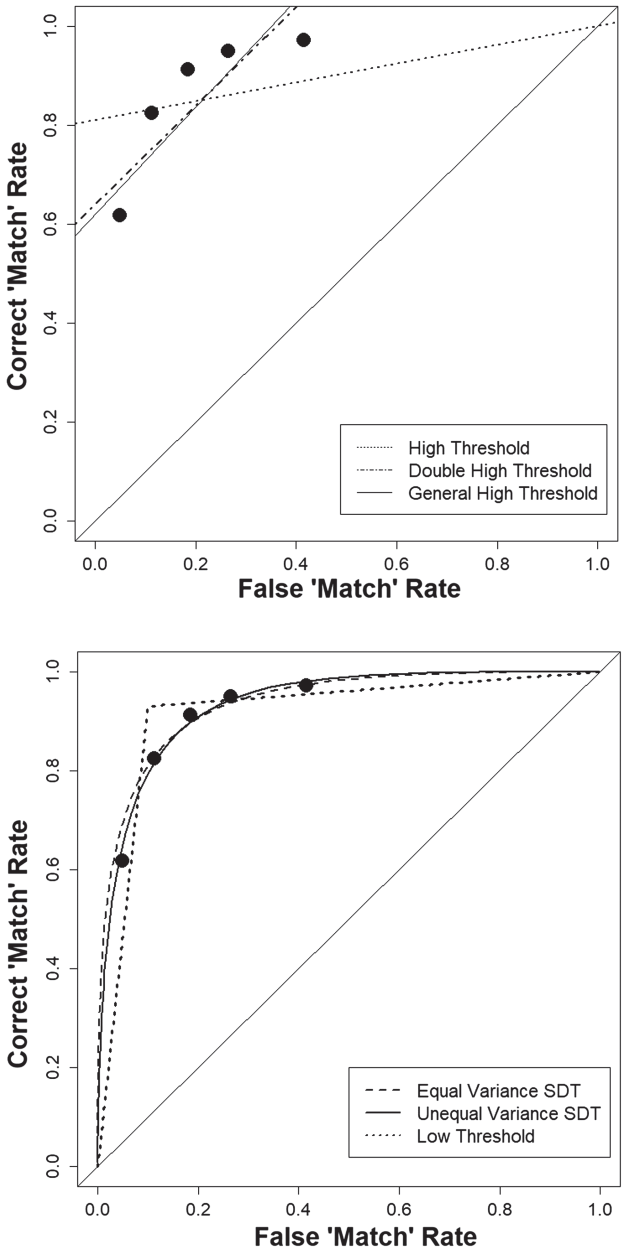
Table 2
Experiment 1b: Best-Fitting Parameters for Each Model

Best-fitting parameters									
Binomial									
p_{h1}	p_{f1}	p_{h2}	p_{f2}	p_{h3}	p_{f3}	p_{h4}	p_{f4}	p_{h5}	p_{f5}
0.62	0.05	0.82	0.11	0.91	0.18	0.95	0.26	0.97	0.41
Unequal variance signal detection									
d	σ	c_1	c_2	c_3	c_4	c_5			
1.98	0.85	1.41	1.21	1.07	0.93	0.65			
Low threshold									
d_s	d_n	b_1	b_2	b_3	b_4	b_5			
0.93	0.10	−0.34	−0.10	0.08	0.18	0.35			
General high threshold									
d_s	d_n	g_1	g_2	g_3	g_4	g_5			
0.62	0.65	0.11	0.41	0.68	0.84	0.94			
High threshold									
d	g_1	g_2	g_3	g_4	g_5				
0.81	0.04	0.11	0.19	0.29	0.45				
Double high threshold									
d	g_1	g_2	g_3	g_4	g_5				
0.64	0.10	0.39	0.66	0.83	0.94				
Equal variance signal detection									
d	c_1	c_2	c_3	c_4	c_5				
2.16	1.81	1.22	0.86	0.59	0.22				

Note. The σ parameter reflects the standard deviation of the match distribution.

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Figure 14
ROC Curves in Experiment 1b



Note. Experiment 1b: Top panel: Empirical ROC points of hit rates as a function of false alarm (black dots) and predicted ROC curves for the high-threshold, double high-threshold, and general high-threshold models. Bottom panel: Empirical ROC points of hit rates as a function of false alarm (black dots) along with predicted ROC curves for the low-threshold, unequal variance signal detection, and equal variance signal detection models. ROC = receiver operating characteristics; SDT = signal detection theory.

and the equal variance SDT models are superior to the alternative models. Taken together, these results provide strong evidence for the success of the UVSD model in accounting for performance in the GFMT test.

Discussion

The results of Experiments 1a and 1b provide considerable support in the unified confidence–similarity UVSD model of face matching. First, the confidence–accuracy correspondence predicted by this model was observed. Second, the UVSD outperformed all the other psychological models (excepting the binomial model, which is not a genuine psychological model), and producing (along with the equal variance SDT model) the best-fitting ROC curve. Taken together, these results strongly support the UVSD, giving currency to the notion that face-matching performance is affected by both perceptual and decisional components and that the internal representation that guides face matching is continuous rather than discrete. When matching faces, participants evoke a continuous signal, one that can be modeled as an identity-signal; participants respond by placing a response criterion, which also determines their confidence in their judgment. As such, the UVSD provides a useful theoretical framework for understanding how confidence judgments are held in face matching. The UVSD can also account for the absence of a match–mismatch correlation. This can simply be understood by adhering to the assumption that in the face-matching task, participants are not representing individual face items, but rather face pairs. Hence, a given face item is always affected by the context of the other face with which it is paired and therefore may evoke different (uncorrelated) identity signals in match and mismatch face pairs.

Experiment 2

At this stage, it is important to show that the UVSD can account for data collected with other sets of faces. In the current experiment, the unified confidence–similarity UVSD model is subjected to scrutiny using another standard face-matching test—the KFMT (Fysh & Bindemann, 2018). In Experiment 2a, participants are tested in the original version of the KFMT, which requires binary-matching responses, while in Experiment 2b, participants perform in a modified version of the KFMT, in which they are asked to rate their confidence in their matching decisions. Central aspects of the proposed UVSD are tested using the same statistical methods as in Experiment 1.

Method

Participants

A total of 124 participants were recruited from Ariel University pool of participants. They were compensated with course credit. Sixty participants ($F = 43$, $M = 17$, $M_{age} = 24.42$, $SD = 6.03$) took part in Experiment 2a, while sixty four ($F = 33$, $M = 31$, $M_{age} = 27.3$, $SD = 11.79$) took part in Experiment 2b.

Stimuli and Apparatus

The stimuli in Experiments 2a and 2b consisted of 40 pairs of unfamiliar faces. These faces comprise the short version of the KFMT (Fysh & Bindemann, 2018). KFMT is a standard test specifically designed for the assessment of unfamiliar face matching. Each face pair stimulus was consisted of two images placed over white background: a frontal photo on the right and a student ID photo on the left. The photo on the right was a color high-quality digital image measuring 283 × 332 pixels of a person taken under controlled

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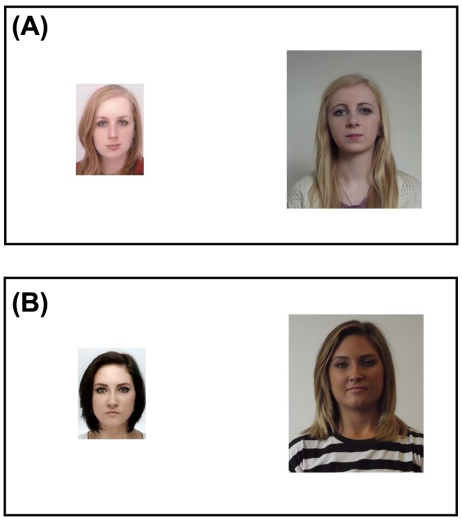
lighting with neutral pose and expression. The photo on the left was an ID color image measuring 142 × 192 pixels. The ID images were not taken under laboratory restricted conditions and therefore varied with respect to pose, expression, and lighting. An example of a face pair from the KFMT is illustrated in Figure 15. Half of the face pairs (20) were identity-match pairs, while the other half (20) were identity-mismatch pairs. The face pairs in the KFMT are constructed such that the same identity (out of two) that shows up in a match face pair appears in a corresponding mismatch face pair. This setup of stimuli allows one to assess the relations between match and mismatch trials at the item level.

Procedure and Design

Experiments 2a and 2b were both conducted online using *Qualtrics* software. Stimuli were presented randomly, one at a time. Participants in Experiment 2a engaged in a binary-choice task. On each trial, they were presented with a face pair, randomly picked from the set by the computer, and were asked: "Do the two faces depict the same person or belong to different people?" Participants responded by clicking one of two buttons posited below the pictures. One button was underlined with the label "same," and the other with the label "different." Participants in Experiment 2b engaged in a confidence-rating task with the same set of face-pairs. On each trial, participants were presented with a randomly chosen face pair and the following question: "Do the two faces depict the same person or belong to different people? Please indicate your confidence in your answer." Participants responded by clicking one of six possible choice buttons. The choices were placed along a horizontal line from left to right starting with 1 = *different high confidence*, 2 = *different medium confidence*, 3 = *different low confidence*, 4 = *same low confidence*, 5 = *same medium confidence*, and 6 = *same high confidence*.

Figure 15

Experiments 2a and 2b: Examples of Face Pairs From the KFMT (Fysh & Bindemann, 2018)



Note. Panel A: The faces mismatch. Panel B: The faces match. See the online article for the color version of this figure.

Results

Experiment 2a

Overall accuracy in the experiment, as measured by probability correct $p(c)$, amounted to 0.63 ($SD = 0.48$). Hit and false-alarm rates amounted to 0.54 and 0.26, respectively. These are in agreement with the published data of the KFMT (Fysh & Bindemann, 2018). Match face pairs (0.54, $SD = 0.014$) were significantly less accurate than mismatch face pairs (0.73, $SD = 0.012$), $t(59) = 5.01$, $p < .0001$. Previous studies have documented either weak negative correlation or no positive correlation between match and mismatch performance, a pattern which arguably supports the dissociation between match and mismatch trials (Megreya & Burton, 2007). To test whether this pattern replicates in the current experiment, I computed the correlation between performance in match face pairs and mismatch face pairs at the *item level*. The Pearson correlation coefficient was close to zero, $C_p = -0.091$, $t(18) = -0.38$, $p = .79$. At the observer level, a medium negative correlation has been observed, $C_p = -0.45$, $t(58) = -3.92$, $p = .0001$. These results confirm earlier observations (Megreya & Burton, 2007).

Experiment 2b

Overall accuracy rate, as measured by probability correct $p(c)$, amounted to 0.62 ($SD = 0.48$), which is comparable to that observed in Experiment 1a ($t < 1$). This suggests that shifting from a binary response to a confidence-rating task has not altered the primary characteristics of the face-matching task. The rates of probability correct $p(c)$ in match trials (0.67, $SD = 0.17$) and in mismatch trials (0.56, $SD = 0.20$) were comparable, $t(19) = 1.89$, $p = .07$. Next, I assessed the relations between match and mismatch performance. The Pearson correlation between performance in match and mismatch trials at the *item level* was close to zero, $C_p = -0.06$, $t(18) = 0.28$, $p = .77$. At the observer level, the Pearson correlation coefficient was medium in size and negative, $C_p = -0.39$, $t(62) = -3.43$, $p = .005$. These results are comparable with those observed in the literature (Megreya & Burton, 2007) and with those documented in Experiments 1a and 1b; they also suggest that switching from a binary-matching task to a confidence-rating task did not alter the main characteristics of the data.

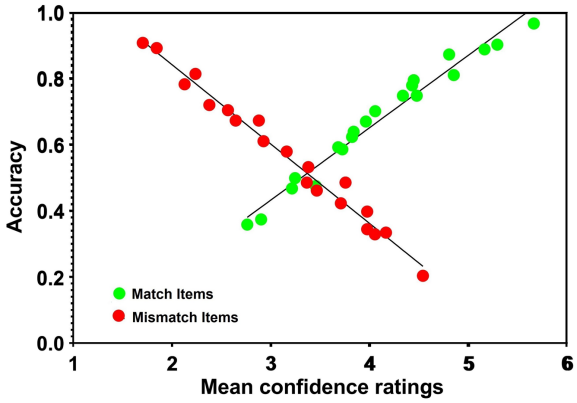
Confidence and Accuracy

Mean accuracy rates and mean confidence ratings (on a scale of 1–6) were computed separately for each face pair item across all observers. Figure 16 plots mean accuracy as a function of average confidence ratings. This type of plot is often dubbed a confidence–accuracy plot (Weber & Brewer, 2004). It gives the probability that a response is correct at a given level of confidence. As can be noted, the pattern observed is the one predicted by the UVSD (see Figure 7). For both match, $R^2 = 0.96$, $t(18) = 23.65$, $p < .001$, and mismatch, $R^2 = 0.98$, $t(18) = -30.61$, $p < .001$, face pairs, the accuracy–confidence correspondence exhibited strong positive linear relations. For both match and mismatch face pairs, accuracy improved as confidence ratings increased. The 1–6 ratings scale dictates a positive slope for match stimuli and a negative slope for mismatch stimuli in the exact manner predicted by the unified confidence–similarity UVSD model. Critically, these patterns are inconsistent with a *positive–negative asymmetry*, which entails linear confidence–accuracy correlation

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Figure 16

Experiment 2b: Confidence–Accuracy Plots Showing Mean Accuracy as a Function of Mean Confidence Ratings for Match and Mismatch Trials at the Item Level



Note. Confidence ratings 1–6 are as follows: 1 = *mismatch high confidence*, 2 = *mismatch medium confidence*, 3 = *mismatch low confidence*, 4 = *match low confidence*, 5 = *match medium confidence*, and 6 = *match high confidence*. See the online article for the color version of this figure.

for positive decisions ("match") and weak or no relations for negative decisions ("mismatch"; Brewer & Wells, 2006; Sporer et al., 1995; Weber & Brewer, 2004). The current results are consistent with symmetric accuracy–confidence relations for match and mismatch trials, and they replicate results from an earlier study by Stephens et al. (2017). The confidence results are important because they support the notion that match and mismatch trials are governed by a single mechanism. This mechanism is correctly characterized by the UVSD.

Fitting SDT Models

Table 3 summarizes the results of the model-fitting procedure, and Table 4 gives the best-fitting parameters per model. As can be noted, the binomial model (−LL = 7,428) produced the highest likelihood value. This is well expected, given that it has the largest number of free parameters. The next best-fitting models were the UVSD (−LL = 7,429), the equal variance SDT (−LL = 7,429), and the low-threshold model (−LL = 7,435). The general high-threshold model (−LL = 7,440), double high-threshold model (−LL = 7,440), and the high-threshold model (−LL = 7,492) followed. The log-likelihood values reveal that the UVSD and the equal variance SDT models provided better fit to the data than the other discrete models.

Table 3

Experiment 2b: Summary of Fitting Results

Model	Parameters	−LL	AIC
Binomial	10	7,428	14,876
General high threshold	7	7,440	14,894
Low threshold	7	7,435	14,884
Unequal variance SDT	7	7,429	14,872
High threshold	6	7,492	14,996
Double high threshold	6	7,440	14,892
Equal variance SDT	6	7,429	14,846

Note. −LL = −log likelihood; AIC = Akaike information criterion; SDT = signal detection theory.

Figure 17 presents the pertinent models in their hierarchically nested form. This figure makes explicit the relations between a model's complexity and its location in the hierarchy. In order to find the most appropriate model, it is customary to perform statistical comparisons between general and restricted models using the G^2 statistics (Ashby & Lee, 1991; Fitousi, 2014; Thomas, 2001). Recall that we fail to reject a restricted model if G^2 is not significant. In this test, the binomial model could be rejected in favor of the restricted UVSD, $\chi^2(3) = 2$, $p = .57$, and the general high threshold was rejected in favor of the double high-threshold, $\chi^2(1) = 0$, $p = 1$. In addition, the UVSD could be rejected in favor of the equal variance SDT, $\chi^2(1) = 0$, $p = 1$. All the other comparisons yielded significant results, which supported the general models. This entails that the winning models of these set of tests are the equal variance SDT and the UVSD. In that case, the former model has less parameters than the latter and is therefore preferable. In addition, both the UVSD and the equal variance SDT models reveal that the standard deviation of the match distribution (indicated by the parameter σ in Table 4) and the standard deviation of the mismatch distribution are roughly equal ($\sigma_{match} = 0.98$ while $\sigma_{mismatch} = 1$). This is not unexpected, given my assumption that experimenters either tacitly or deliberately dictate the variances of the match and mismatch distributions by the mere act of pairing the images in a certain way. Face pairs in the KFMT were presumably constructed with the intention of equating the difficulty of match and mismatch trials. Next, I provide another rigorous test of the UVSD and its competitors by assessing the ROC curves produced by each model.

ROC Curves

The empirical shape of the ROC curve is highly diagnostic and can potentially help researchers decide between competing models (Mickes et al., 2017; Wixted & Mickes, 2014; Yonelinas, 1994). The high-threshold, double high-threshold, and general high-threshold models predict straight isosensitivity lines, whereas the equal and unequal variance SDT models predict curvilinear lines; in addition, the low threshold predicts a two-limb pattern (Green & Swets, 1966; Macmillan & Creelman, 2004). I extracted the best-fitting parameters for each model (see Table 4) and used them to generate the best-fitting ROC curve for each model by plugging them into their corresponding equations. Figure 18 presents the modeled lines along with the empirical ROC points. As can be noted, the points of the empirical ROC are most consistent with a curvilinear shape. Models that predict linear isosensitivity lines such as the high-threshold, double high-threshold, and general high-threshold models are inadequate (see top panel of Figure 18). The high-threshold model gives the worst fit to the ROC points, which is expected given that it produced the lowest likelihood of the three models. The double high-threshold and the general high-threshold models yielded two quite similar isosensitivity lines, but none of them were capable of accommodating the curvilinear shape of the empirical ROC. The bottom panel of Figure 18 presents the predicted ROC curves for the low-threshold model, along with those for the equal and unequal SDT models. A visual inspection suggests that the equal variance SDT and UVSD models produced the

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Table 4
Experiment 2b: Best-Fitting Parameters for Each Model

Best-fitting parameters									
Binomial									
p_{h1}	p_{f1}	p_{h2}	p_{f2}	p_{h3}	p_{f3}	p_{h4}	p_{f4}	p_{h5}	p_{f5}
0.41	0.12	0.55	0.28	0.67	0.35	0.78	0.43	0.89	0.59
Unequal variance signal detection									
d	σ	c_1	c_2	c_3	c_4	c_5			
0.57	0.98	1.12	0.52	0.14	−0.13	−0.56			
Low threshold									
d_s	d_n	b_1	b_2	b_3	b_4	b_5			
0.70	0.42	−0.62	−0.27	−0.02	0.21	0.52			
General high threshold									
d_s	d_n	g_1	g_2	g_3	g_4	g_5			
0.18	0.19	0.15	0.38	0.57	0.69	0.86			
High threshold									
d	g_1	g_2	g_3	g_4	g_5				
0.28	0.10	0.30	0.47	0.59	0.75				
Double high threshold									
d	g_1	g_2	g_3	g_4	g_5				
0.18	0.14	0.38	0.56	0.69	0.85				
Equal variance signal detection									
d	c_1	c_2	c_3	c_4	c_5				
0.57	1.13	0.53	0.14	−0.13	−0.57				

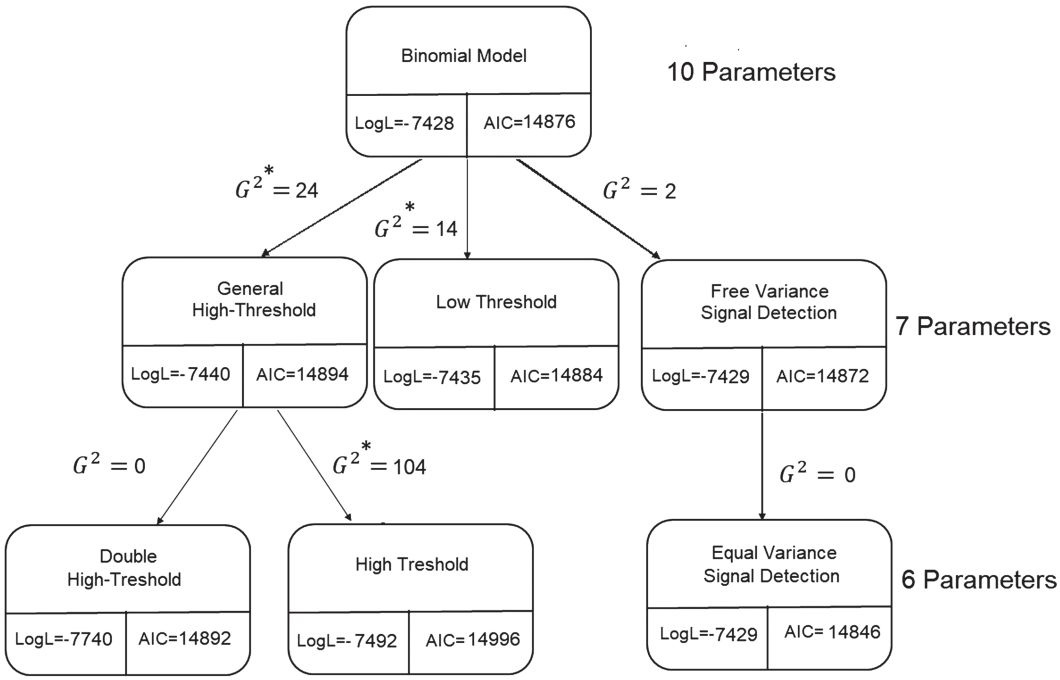
Note. The σ parameter reflects the standard deviation of the match distribution.

best-fitting ROC curves. This result is consistent with the fitting results. The low-threshold model was close, but could not produce a curvilinear line due to its two-limb shape. In summary, consistent with the current confidence–similarity UVSD model, both the UVSD and equal variance SDT models have shown the best-fitting ROC curves.

Discussion

Experiments 2a and 2b provided another rigorous test of the unified confidence–similarity UVSD model of face matching. First, the experiments demonstrated that switching from a choice task to a confidence-rating task did not alter central aspects of performance.

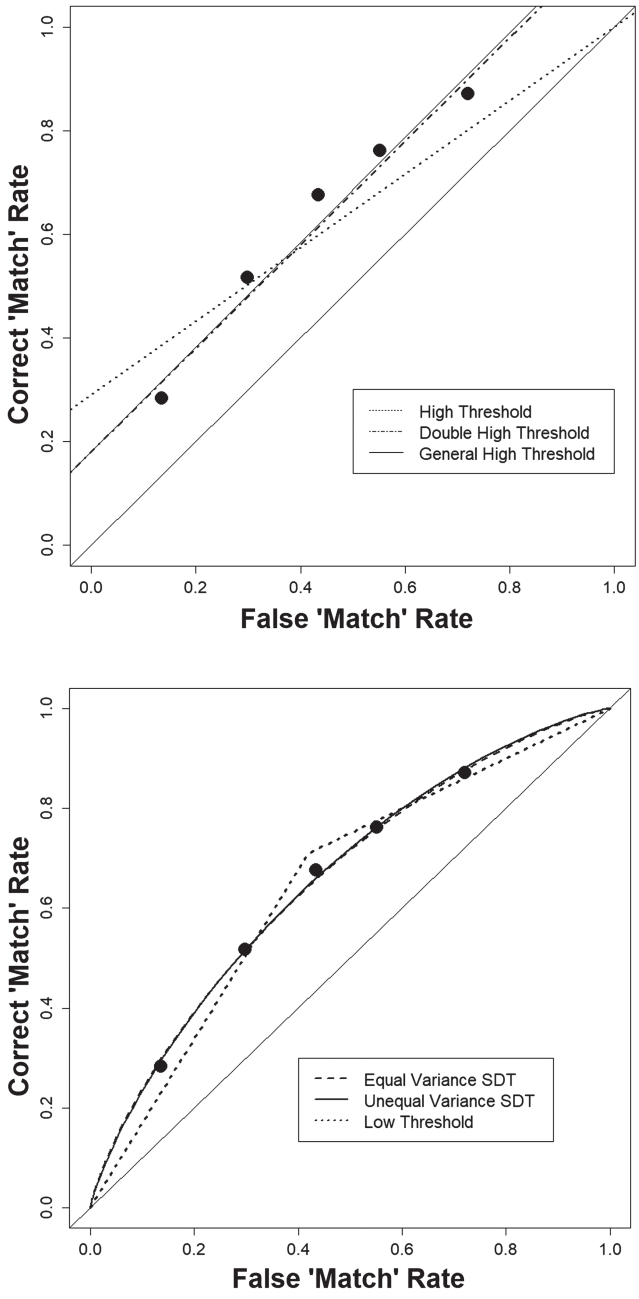
Figure 17
Experiment 2b: A Hierarchy of Seven Models for the Confidence Rating Results



Note. Included are model comparison statistics G^2, LL, and AIC. LL = log likelihood; AIC = Akaike information criterion.
$^{*}p < .05$.

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Figure 18
ROC Curves for Experiment 2b



Note. Experiment 2b: Top panel: Empirical ROC points of hit rates as a function of false alarm (black dots) and predicted ROC curves for the high-threshold, double high-threshold, and general high-threshold models. Bottom panel: Empirical ROC points of hit rates as a function of false alarm (black dots) along with predicted ROC curves for the low-threshold, unequal variance SDT, and equal variance SDT models. ROC = receiver operating characteristics; SDT = signal detection theory.

Most importantly, the experiments reveal that the UVSD and its special case model—the equal variance SDT model—outperformed all other competitors, fitting the data better than any of the alternatives. They also accommodated the observed curvilinear ROC curve very well. The results suggest that in the KFMT, the variances of the match (within-identities) and mismatch (between-identities) distributions are roughly equal. In addition, the data exhibited the exact confidence–accuracy pattern predicted by the UVSD. In particular, the confidence–accuracy plots revealed a symmetric positive–negative accuracy–confidence correspondence, a pattern that is fully consistent with the predictions of the UVSD. These findings have several important theoretical implications for face matching: (a) They support the notion that face-matching performance is affected by both perceptual and decisional components, (b) they give currency to the notion that face matching is governed by continuous rather than discrete representations, (c) they corroborate the notion that match and mismatch decisions are held by a single mechanism, and (d) they validate the predictions of the UVSD model concerning confidence and accuracy. One issue that remains to be tested is the predictions of the unified confidence–similarity UVSD model with respect to similarity judgments. This issue is addressed in Experiment 3.

Experiment 3

The purpose of Experiment 3 was to test the assumptions of the unified confidence–similarity UVSD model with respect to similarity. Evidently, the task of matching unfamiliar faces is governed by an image-based process that relies on the pictorial aspects of the images (Burton, 2013; Jenkins & Burton, 2011). As such, it requires observers to take advantage of the visual input from the two photos. The degree of perceived similarity/dissimilarity of a given face pair has been shown to be a good predictor of matching accuracy (Fysh & Bindemann, 2023). Although similarity is considered to be a slippery concept (Tversky, 1977), it is essential for understanding how people recognize and match faces. The UVSD offers a simple but powerful approach to modeling similarity in the face-matching task. First, the decision variable in the UVSD is a similarity signal, and thus perceived similarity of a given face pair should be related to its strength in the UVSD space. Second, the UVSD adopts a fairly straightforward distinction between global and local similarity. Global similarity is an aspect of the entire set of face pairs in an experiment, and as such, it can be modeled across experiments or stimuli sets, using the discriminability parameter. Local similarity, in contrast, is an aspect of a given face pair. It has been found that highly similar match face pairs produce higher hit rates than less similar match face pairs, while highly similar mismatch face pairs yield higher rates of false alarms than dissimilar mismatch pairs (Fysh & Bindemann, 2023). The unified confidence–similarity UVSD model can explain this pattern by assuming that similarity ratings are evoked by the observer in the same way confidence ratings are generated, namely by placing a set of response criteria. Each face pair evokes a similarity signal that falls within an area bounded by the response criteria. Its location determines its similarity rating. Under the assumption that the face space remains invariant across tasks, it is expected that the response criteria for the similarity and confidence tasks are highly correlated (but not necessarily identical). This implies that (a) perceived similarity of a face pair is positively correlated with its confidence rating and (b) perceived similarity of a face pair predicts its accuracy in the confidence-rating matching task. According to this framework, for match face pairs, accuracy should increase as a function of similarity, while for mismatch face pairs, accuracy should decrease as similarity increases. The goal of Experiment 3 is to test these predictions.

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Method

Participants

A total of 40 participants participated ($F = 31$, $M = 9$, $M_{age} = 23.4$, $SD = 2.44$). Participants were recruited from Ariel University pool of participants and were compensated with a course credit.

Stimuli and Apparatus

The exact GFMT stimuli from Experiments 1a and 1b were used in Experiment 3.

Procedure and Design

The experiment was conducted in the lab. Stimuli were presented randomly, one at a time. Participants performed in a similarity-rating task. They were asked, "To what extent are the two faces similar?" Participants responded by clicking one of six possible choices. The choices were placed along a horizontal line from left to right starting with 1 = *high dissimilarity*, 2 = *medium dissimilarity*, 3 = *low dissimilarity*, 4 = *low similarity*, 5 = *medium similarity*, and 6 = *high similarity*. One of the goals of the experiment is to relate similarity ratings to confidence ratings for the same set of face pairs. I have judiciously designed this experiment to collect only similarity ratings, without asking participants to give additional confidence ratings. The rational being to prevent any possible interactions between similarity and confidence decisions. Confidence ratings for the same face pairs were already collected in Experiment 1b with a different sample of participants.

Results

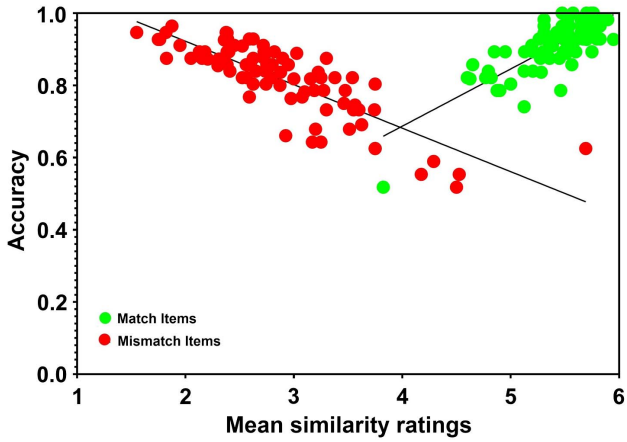
Overall mean similarity ratings for match trials (5.4, $SD = 0.36$) was higher than for mismatch trials (2.87, $SD = 0.7$), $t(83) = 29.72$, $p < .001$, indicating that images of the same facial identity were perceived as more similar than images of different facial identities. This outcome is consistent with the UVSD assumption that face matching is governed by a similarity signal, one that can be possibly stronger for match than mismatch face pairs. Moreover, the variability of similarity ratings was higher in the between-identitiesy (mismatch) than in the within-identities (match) condition, $F(83, 83) = 0.26$, $p < .0001$. The UVSD model predicts that the variability of perceived similarity ratings should be related to the variability of the theoretical continuous identity–similarity signal. The results are in line with outcomes from the matching task in Experiment 1b, which pointed to higher variance in the mismatch distribution than in the match distribution. It is interesting to note that Kramer et al. (2018) have shown comparable results with automatic systems tested on the same GFMT face pairs deployed here. These authors found that the Euclidean distances between the constituent images of face pairs in the PCA + Linear Discriminant Analysis (LDA) face space were, on average, shorter for same-identity (match) than for different-identity (mismatch) face pairs. Shorter Euclidean distances imply higher degrees of similarity. Their model was capable of separating match and mismatch face pairs, but not perfectly, because there was a clear overlap in the distributions.

Similarity and Accuracy

The unified confidence–similarity UVSD model predicts that accuracy in the face-matching task will be tightly associated with the degree of similarity/dissimilarity of faces in a given face pair. To test this hypothesis, mean similarity ratings were computed for each of the GFMT face pairs. In addition, mean accuracy was computed for each of the GFMT face pair items in the data of Experiment 1b. Figure 19 plots mean accuracy as a function of mean similarity ratings. As can be noted, accuracy with both matches, $R^2 = 0.61$, $F(1, 82) = 129.1$, $p < .0001$, and mismatch, $R^2 = 0.68$, $F(1, 82) = 175.7$, $p < .0001$, face pairs was strongly predicted by their similarity ratings. This pattern is consistent with the predictions of the unified confidence–similarity UVSD model. These results provide solid evidence that matching accuracy of a given face pair is strongly associated with the degree of perceived similarity/dissimilarity of this face pair. According to the unified confidence–similarity UVSD model, the similarity signal evoked by a face pair along with a set of response criteria determine this pair's degree of judged similarity. Face pairs that are located within the bounds of conservative response biases are (a) perceived as more similar/dissimilar, (b) received higher confidence ratings, and (c) matched with higher accuracy. This finding is remarkable, given that the accuracy rates and similarity ratings were collected from two different groups of participants. This implies the presence of considerable agreement across participants on the degree of similarity/dissimilarity of face pairs to the effect that one group's similarity judgment can predict another group's accuracy. The observed pattern is in agreement with recent findings by Fysh and Bindemann (2023), who demonstrated comparable relations in a within-participant design. The results provide considerable support in the confidence–similarity UVSD model, which offers a parsimonious account of the observed pattern by virtue of the assumption that a single mechanism exists for confidence- and similarity-rating tasks.

Figure 19

Experiment 3: Mean Accuracy as a Function of Mean Similarity Ratings per Item for Match and Mismatch Face Pairs



Note. The scale for similarity consisted of following labels: 1 = *high dissimilarity*, 2 = *medium dissimilarity*, 3 = *low dissimilarity*, 4 = *low similarity*, 5 = *medium similarity*, and 6 = *high similarity*. See the online article for the color version of this figure.

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Similarity and Confidence

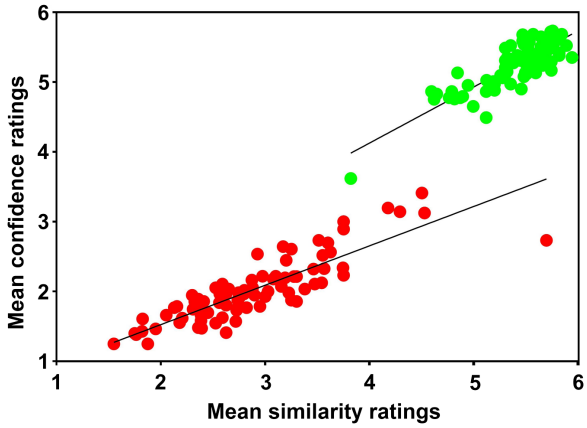
The unified confidence–similarity UVSD model of face matching postulates that similarity and confidence are two related aspects of face-matching performance. Participants in the confidence task and participants in the similarity task operate on the same face space by evoking a similarity signal and by placing the response criteria. The response criteria for the similarity and confidence tasks are not necessarily identical, but are highly correlated. Arguably, the two are generated by the same psychological mechanism. Thus, a direct prediction of the UVSD model is that similarity ratings will be positively correlated with confidence ratings. To test this hypothesis, I computed the average similarity ratings for each of the GMFT face pairs, along with the mean confidence ratings of each of the GMFT face pairs in the data of Experiment 1b. Recall, that the two experiments were conducted with different groups of participants, one group provided confidence ratings, while the other gave similarity ratings for the same set of face pairs. Figure 20 presents the relations between mean similarity ratings and mean confidence ratings. As can be noted, similarity ratings with both match, $R^2 = 0.70$, $F(1, 82) = 194.1$, $p < .0001$, and mismatch, $R^2 = 0.73$, $F(1, 82) = 224.3$, $p < .0001$, face pairs strongly predicted their confidence ratings. It is remarkable that such positive high correlations between the two variables emerge, although they were collected from two independent groups of participants. This implies that similarity ratings of one group of participants can predict the confidence ratings of another group. More importantly, the degree of similarity/dissimilarity of a face pair can predict the degree to which the participant is confident in their matching decision. These results are completely in agreement with the predictions of the UVSD and, as such, provide considerable support in the model.

Fitting Models

The finding of a tight correspondence between similarity ratings and confidence ratings is astounding. It is commensurate with the predictions of the UVSD, suggesting that similarity and confidence are governed by related mechanisms of placing the response criteria. As noted earlier, observers may place their response criteria at the exact same locations in both the similarity and confidence tasks; alternatively, the participants may use two sets of criteria, one for each task, but these criteria are highly correlated by virtue of their ordered nature and the invariance of the underlying face space. One way of testing this conjecture is by making the assumption that similarity ratings *are* confidence ratings and then by fitting the candidate models to the data from the similarity-rating task. This also allows us to test the validity of the UVSD. The similarity ratings, which span on a 1–6 scale, were converted to confidence ratings along a 1–6 scale using a simple one-to-one correspondence as follows: 1 = *high dissimilarity* → *high-confidence mismatch*, 2 = *medium dissimilarity* → *medium-confidence mismatch*, 3 = *low dissimilarity* → *low-confidence mismatch*, 4 = *low similarity* → *low-confidence match*, 5 = *medium similarity* = *medium-confidence match*, and 6 = *high similarity* → *high-confidence match*. The conversion makes sense because confidence and similarity are intertwined in the decision process. For example, a face pair that is rated as highly dissimilar should also lead to high-confidence mismatch decision, while a face pair that is rated as highly similar should likely be accompanied by a high-confidence match decision. Along this vein, low-similarity or low-dissimilarity decisions should likely be accompanied by, respectively, low-confidence mismatch and match decisions.

Figure 20

Experiment 3: Mean Confidence Ratings (Experiment 1b) as a Function of Mean Similarity Ratings per Item for Match and Mismatch Face Pairs



Note. The scale for similarity consisted of following labels: 1 = *high dissimilarity*, 2 = *medium dissimilarity*, 3 = *low dissimilarity*, 4 = *low similarity*, 5 = *medium similarity*, and 6 = *high similarity*. See the online article for the color version of this figure.

The results of this approach provided excellent support in the similarity–confidence hypothesis and in the validity of the UVSD. Table 5 presents the values of the model-fitting procedure. As expected, the most general model, the binomial model (−LL = 14,496), exhibited the highest likelihood value. The UVSD model came second (−LL = 14,506) and the equal variance SDT (−LL = 14,508) came third. The low threshold (−LL = 14,583) was fourth. The high-threshold models came last.

Judged by their AIC values, the equal variance SDT (AIC = 29,004), the binomial (AIC = 29,012), and the UVSD (AIC = 29,026) provided the best fit. Model comparisons between general and nested models using the G^2 statistics revealed that all the G^2 statistics were significant (see Figure 21), suggesting that none of the general models could not be rejected in favor of their restricted versions. However, given that the binomial model is not a psychologically driven model, the UVSD is the next best model in the hierarchy (with the lowest G^2 value) coming after the binomial

Table 5

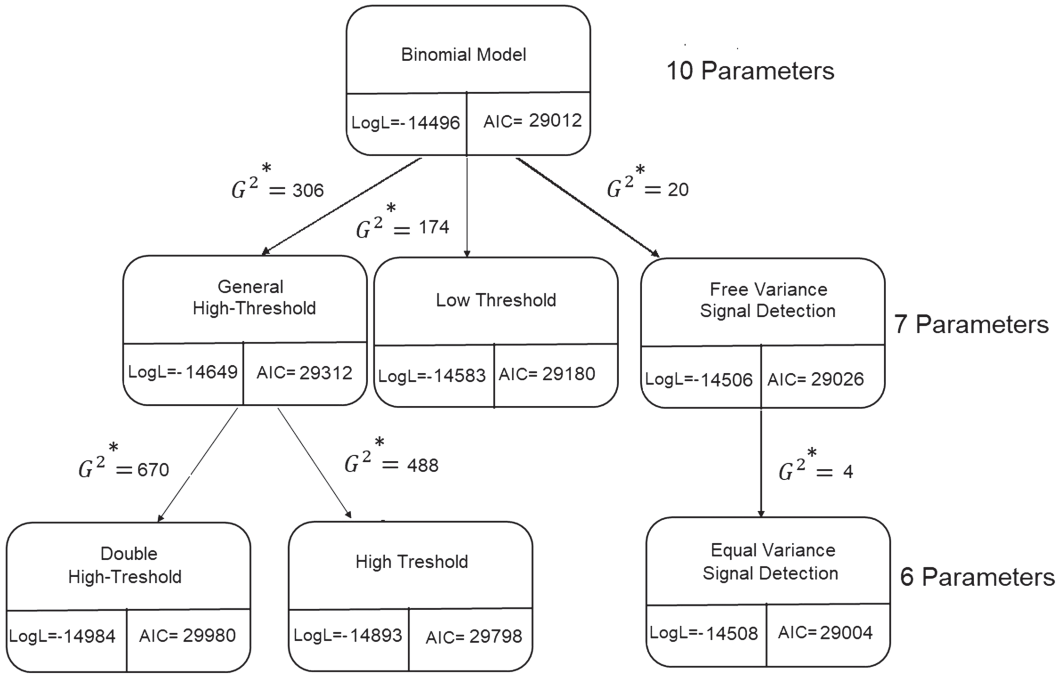
Experiment 3: Summary of Fitting Results Under the Assumption That Similarity Ratings Reflect Confidence Ratings

Model	Parameters	−LL	AIC
Binomial	10	14,496	29,012
General high threshold	7	14,649	29,312
Low threshold	7	14,583	29,180
Unequal variance SDT	7	14,506	29,026
High threshold	6	14,893	29,798
Double high threshold	6	14,984	29,980
Equal variance SDT	6	14,508	29,004

Note. −LL = −log likelihood; AIC = Akaike information criterion; SDT = signal detection theory.

Figure 21

Experiment 3: A Hierarchy of Seven Models for the Confidence Rating Results Based on Similarity Ratings Under the Assumption That They Reflect Confidence Ratings



Note. Included are model comparison statistics G^2, LL, and AIC. See text for technical details. LL = log likelihood; AIC = Akaike information criterion.

model. These results entail that the UVSD is the most appropriate model also when considered as a part of a hierarchy of models.

ROC Curves

The current approach treats similarity ratings as if they were confidence ratings. This assumption is justified both theoretically and empirically. First, the confidence–similarity model assumes that the two are very close quantitatively. Second, the empirical results of Experiment 3 confirmed this conjecture, revealing a strong and positive correlation between similarity and confidence ratings at the face pairs level. Third, fitting the models to the converted similarity ratings yielded results that were quite comparable to those obtained with confidence ratings in Experiment 1b. In this section, the similarity–confidence analogy is pushed a little bit further. As in the previous experiments with confidence ratings, ROC analyses are performed on the converted similarity ratings. To do so, I extracted the best-fitting parameters for each model (see Table 6) based on the model-fitting procedure performed in the previous section and used those parameters to generate an ROC curve for each of the tested models. The top panel of Figure 22 presents the empirical ROC points along with the best-fitting ROC curves for the high-threshold, general high-threshold, and double high-threshold models. As can be noted, the empirical ROC points cannot completely be ordered along a straight line. Although all the three high-threshold models do a good job in fitting the points, none of them really passes through all the points. Because these models predict a straight ROC line, they basically cannot accommodate the curvilinear shape. The bottom panel of Figure 22 illustrates the empirical ROC points along with the best-fitting ROC curves for the UVSD, the equal variance SDT model, and the low-threshold model. Eyeballing these curves, it becomes apparent that the UVSD and the equal variance SDT provide the best-fitting ROC curves to the data. This is consistent with the results of the model-fitting procedure. Notably, the standard deviation parameter of the match distribution was close to 1 (σ = 1.06), suggesting that within-identities and between-identities variances were comparable. The low-threshold model closely mimicked the ROC points, but due to its two-limb shape could not accommodate the curvilinear shape of the ROC. In summary, the UVSD and its subspecies—the equal variance SDT model—turn out to be the most appropriate models for the similarity data of Experiment 3, under the assumption that similarity ratings are related, in one way or another, to confidence ratings.

Discussion

Experiment 3 was designed to test a set of predictions derived from the UVSD concerning the role of similarity in face matching and the relations between similarity and accuracy, and between similarity and confidence ratings. Participants rated the similarity of the GFMT face pairs (Burton et al., 2010). The results showed that similarity serves as a prominent factor in face matching. First, a strong positive and linear correspondence at the item level (face pairs) was documented between similarity ratings and accuracy rates. For match face pairs, accuracy increased as similarity increased, whereas for mismatch face pairs, accuracy decreased as similarity ratings increased. This is the exact pattern predicted by the unified confidence–similarity UVSD

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Table 6
Experiment 3: Best-Fitting Parameters for Each Model

Best-fitting parameters									
Binomial									
p_{h1}	p_{f1}	p_{h2}	p_{f2}	p_{h3}	p_{f3}	p_{h4}	$_{pf4}$	p_{h5}	p_{f5}
0.68	0.1	0.85	0.20	0.94	0.37	0.97	0.55	0.98	0.69
Unequal variance signal detection									
d	σ	c_1	c_2	c_3	c_4	c_5			
1.87	1.06	1.33	0.80	0.30	−0.12	−0.47			
Low threshold									
d_s	d_n	b_1	b_2	b_3	b_4	b_5			
0.89	0.13	−0.23	0.067	0.28	0.49	0.65			
General high threshold									
d_s	d_n	g_1	g_2	g_3	g_4	g_5			
0.72	0.33	0.13	0.32	0.62	0.85	0.95			
High threshold									
d	g_1	g_2	g_3	g_4	g_5				
0.80	0.08	0.20	0.39	0.58	0.71				
Double high threshold									
d	g_1	g_2	g_3	g_4	g_5				
0.48	0.25	0.56	0.84	0.95	0.97				
Equal variance signal detection									
d	c_1	c_2	c_3	c_4	c_5				
1.79	1.30	0.80	0.30	−0.11	−0.46				

Note. Fitting is based on the assumption that similarity ratings are confidence ratings. The σ parameter reflects the standard deviation of the match distribution.

model. Second, under the assumption that similarity and confidence ratings are generated by the same or correlated mechanisms, the UVSD and its subspecies—the equal variance SDT—yielded the best fit to the data. All of these provide a considerable evidence in favor of the unified confidence–similarity UVSD model.

General Discussion

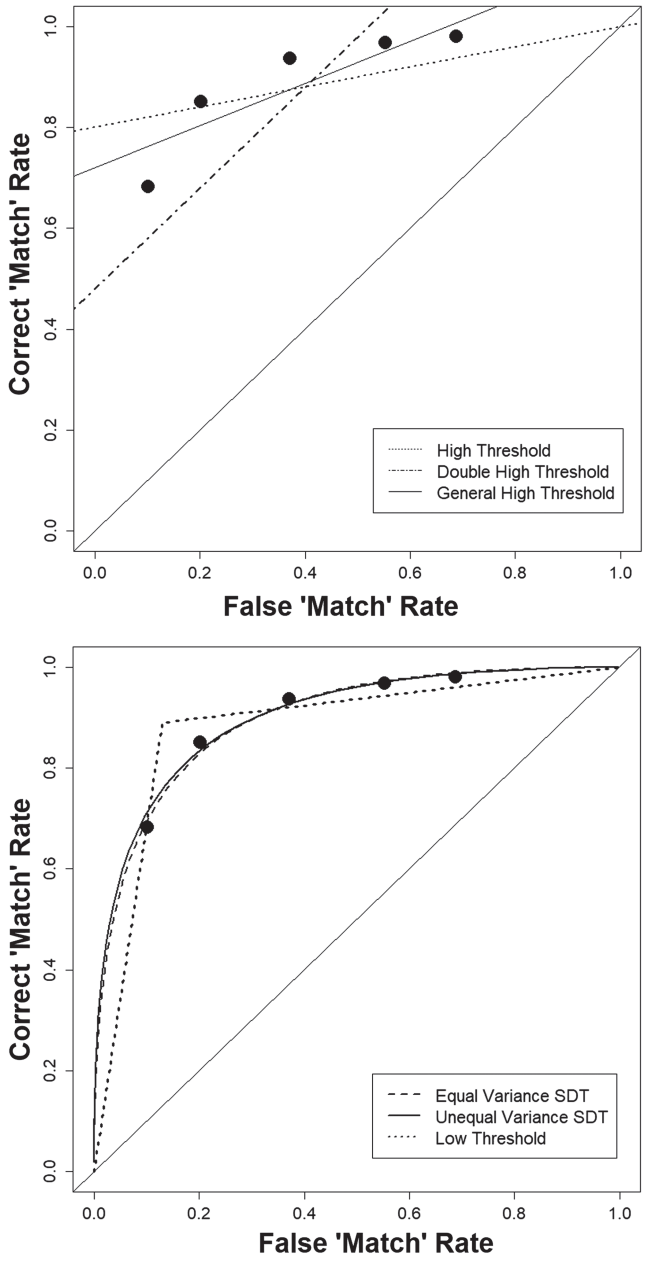
Many researchers hold on to the belief that match and mismatch conditions in the face-matching task are governed by two separate systems (Fysh & Bindemann, 2023; Megreya & Burton, 2006, 2007). The present study offers a more parsimonious account based on a single-system alternative. The framework is dubbed the unified confidence–similarity UVSD model, and it can explain many aspects of performance in the face-matching task by adhering to a single-process mechanism akin to a continuous similarity signal. The model was tested against six alternative competing models in three experiments. The face stimuli were derived from the well-known GFMT (Burton et al., 2010) and the KFMT (Fysh & Bindemann, 2018) face-matching tests. The face stimuli were administrated in choice, confidence-rating, and similarity-rating tasks. Data analyses consisted of hierarchically nested model fitting, ROC curves fitting, and confidence–accuracy plots analyses. All three experiments provided substantial support in the unified confidence–similarity UVSD model. First, in all three experiments, the UVSD or its subspecies, the equal variance SDT, outperformed the alternative models in fitting the data. Most of those competing models postulated discrete rather continuous representations. Thus, the experiments strongly supported the notion that face matching is governed by a continuous rather than discrete signal. This conclusion was also supported by the ROC analyses, in which the UVSD exhibited the best fit. In addition, the predictions of the UVSD with respect to the relations between confidence and accuracy, and between similarity/confidence and accuracy were fully supported. In regard to similarity, in match trials, accuracy increased as perceived similarity increased, whereas in mismatch trials, accuracy decreased as similarity increased. Confidence was positively correlated with similarity for both match and mismatch trials. With respect to confidence, symmetric accuracy–confidence plots were observed, such that accuracy increased as confidence increased for both match and mismatch trials. The UVSD accounted successfully for the complex relations that emerged among all of those variables. The upshot is that the unified confidence–similarity UVSD model can serve now as a viable model of face-matching and related tasks.

The model has several distinctive features that make it a viable framework for interpreting face-matching phenomena. First, the model assumes that match and mismatch distributions have unequal variances, an assumption that allows for separate influences of external factors on the match and mismatch distributions. This feature can explain the finding that certain factors exert selective influence on match (mismatch), but not on mismatch (match) conditions (McCaffery & Burton, 2016; Strathie & McNeill, 2016; Wirth & Carbon, 2017), or the effect of unequal frequencies (Stephens et al., 2017). Second, the model can explain the absence of match and mismatch correlation (Megreya et al., 2011) due to its assumption that observers represent face pairs and not single faces. Third, the model separates between perceptual and decisional aspects of behavior and therefore can account for strategic effects and individual differences that are often documented in face matching (Wirth & Carbon, 2017). Fourth, the model can account for confidence ratings and their relations to accuracy. The model predicted strong confidence–accuracy correspondence, a hypothesis that was corroborated by the empirical results. Fifth, the model correctly predicted the relations between similarity ratings, confidence ratings, and accuracy. And lastly, en route to the main goal, a more general featural face-space model has been developed that can account for the computation

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Figure 22

ROC Curves for Experiment 3



Note. Experiment 3: Top panel: Empirical ROC points of hit rates as a function of false alarm (black dots) and predicted ROC curves for the high-threshold, double high-threshold, and general high-threshold models. Bottom panel: Empirical ROC points of hit rates as a function of false alarm (black dots) along with predicted ROC curves for the low-threshold, unequal variance signal detection, and equal variance signal detection models. Fitting is based on the assumption that similarity ratings are confidence ratings. ROC = receiver operating characteristics; SDT = signal detection theory.

of similarity based on a feature-matching binomial process (Wixted, 2020). The proposed framework, in contrast to most extant geometric face-space models, allows for variability in the representations of the same identity (Jenkins et al., 2011). Thus, the current work sheds new light on the emergence of within-identities and between-identities sources of variance not only in face matching, but also in other face recognition tasks. In summary, the current effort offers, what I believe to be, a broader view of face recognition to address the challenge of image variability that is persistent in many other face recognition tasks (Burton, 2013; Jenkins et al., 2011; White et al., 2022).

UVSD and Neural Mechanisms

Similarity is a fundamental construct in theories of face perception and as such figures prominently in face-space models (Nosofsky, 1988; Trujillo & Anderson, 2023; Valentine, 1991; Vanpaemel & Storms, 2008), where it is construed as distance in some featural or geometrical metric space. Such models have been instrumental in conceptualizing and guiding both psychophysical (Rhodes & Jeffery, 2006) and neurophysiological (Freiwald & Hosoya, 2021) research. For example, in the central prototype model (Valentine, 1991), faces are encoded relative to a normative face, which is located at the center of the space. Distinctive faces occupy areas that are further away from this face and are less densely populated, whereas more typical faces are located closer to the center in areas that are more densely populated. According to this model, similar faces inhabit neighboring areas of space, while dissimilar faces are located further from each other. Similarity also plays a role in the current unified confidences–similarity UVSD model. The model assumes that when a face pair is presented, a similarity signal is automatically computed, with strength that depends on the distance between the faces' representations either on a featural or a General Recognition Theory (Ashby & Townsend, 1986) space. The validity of the UVSD model for face matching is therefore contingent on the validity of the assumption that viewing a pair of faces produces a similarity signal. While the present study has provided substantial behavioral evidence for this conjecture, a question arises whether a plausible biological mechanism exists that can generate such a signal? This also amounts to the question of how does the brain create a face space? I dedicate the next paragraphs to addressing these caveats.

Research in neuroscience has discovered two diametrically opposed neuronal coding systems dedicated to faces. These coding schemes are capable of representing faces as vectors of neuronal activation, which produce face-space representations equivalent to those proposed in cognitive research and in the current work. Such neuronal mechanisms are viable candidates for producing the similarity signal proposed in the current UVSD framework, as well as in related models (Shen et al., 2023). The first neuronal mechanism is dubbed *axis coding* (Freiwald et al., 2009), according to which cells in the middle fundus and middle lateral patches areas are tuned to small subset of features (e.g., relative face width, inter-eye distance). These cells produce ramp-shape tuning curves with maximal responses at or near one feature extreme. When cells are tuned to more than one feature, responses to features are orthogonal. In this coding scheme, the neurons are the axes of the face space, and their joint pattern of firing for each face represents its location in that space. In such a scheme, overall featural similarity between two faces is proportional to the similarity between their firing profiles. The featural similarity model I outlined at the outset bears close conceptual affinity with this biological mechanism. A second mechanism is called *norm-based coding* (Leopold et al., 2006). It was found that neurons in the face-responsive neurons in the

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macaque monkey anterior inferotemporal cortex are tuned to the norm face. Recording from these neurons during the presentation of faces from different regions of the face space revealed V-shape tuning curves with peaks or troughs in the center of face space. Some neurons respond maximally to the average face, whereas others respond minimally. This mechanism gives currency to a norm-based face space (Valentine, 1991), in which distance from the typical face (or similarity) establishes the pattern of neuronal firing. Koyano et al. (2021) have shown that axis coding and norm coding coexist in the same neural populations (Freiwald & Hosoya, 2021). According to these authors, identity coding is accomplished by two subsequent stages. A first, axis-coding scheme is implemented by fast feedforward connections based on stimulus statistics; and a second recurrent system of connections produces norm-based encoding. This model is consistent with a local-to-global dynamics (Fitousi, 2019), in which a face is first represented in detail, and then increasingly attains a more abstract representation.

Of the two well-established neuronal coding schemes, axis coding (Freiwald et al., 2009) seems to be the most suitable candidate for generating the behavioral effects observed in face matching of unfamiliar faces under ambient conditions, as well as for producing the similarity signal postulated in the current UVSD framework. First, face matching with unfamiliar faces is considered by many to be an image-based process (Hancock et al., 2000; Jenkins & Burton, 2011), and as such is expected to rely heavily on pictorial cues and statistical learning. It is likely performed in a bottom-up feedforward manner, consistent with the axis-coding system. Second, the ability of axis coding to represent minute details by coding the constituent features makes it a viable mechanism for producing both within-identities and between-identities variability in the similarity signal. Third, the feasibility of such an operation is validated by the proposed feature-matching model presented at the outset. In this simple model, I provide an existence proof for a binomial feature-matching process that produces a variable similarity signal akin to the Gaussian distributions postulated by the UVSD. Fourth, as noted earlier, the norm-based coding framework (cognitive and biological implementations) ignores ambient image variability in photos of the same facial identity (Burton, 2013; Jenkins et al., 2011). It cannot explain, for example, why a photo of Person A is judged as more similar to a photo of Person B than to another photo of Person A (e.g., violation of the minimality axiom). Thus, norm-based coding is untenable as an explanation of unfamiliar face matching under ambient noise, at least at an early stage, where detailed representations are used. This conclusion is also supported by the robust finding (also confirmed in the present study) that a mirror effect in face matching with unfamiliar faces is absent (Megreya & Burton, 2007). This finding is incommensurate with predictions of norm-based coding, which postulates that the activation of faces should differ due to their location in the face space. The approach predicts, for example, that a distinctive face that is accurately judged as "match" should also be accurately rejected as "mismatch," a phenomenon that should have resulted in a positive correlation at the item level. This prediction has not been corroborated by the available empirical data (Megreya & Burton, 2007). In contrast, the axis-coding system can account for the absence of distinctiveness effects because it does not encode faces with respect to a norm.

UVSD and Automatic Systems Performance

Recent advances in computer vision (Phillips & O'Toole, 2014) using deep convolution neuronal networks (CNNs) have led to highly accurate models of face perception, such as the Visual Geometry Group from Oxford-16 layers depth (Parkhi et al., 2015; Simonyan & Zisserman, 2014) for face recognition and the Siamese (Song et al., 2019) for face matching. This success has likely contributed to the increase in popularity of machine learning (ML) systems of face recognition in forensic, surveillance, and other practical settings (Phillips et al., 2018; White et al., 2015). For example, the VGG-face model (Simonyan & Zisserman, 2014) was designed to recognize faces in variable illumination. It was tested against several benchmarks (Phillips and O'Toole, 2014) and exhibited impressive accuracy levels. A question that arises is how does the present simple UVSD model compare to those state-of-the-art face recognition/match models?

First, it is important to note that although algorithms have become more efficient, humans continue to be indispensable for most face recognition tasks, working alone or in collaboration with machines. The presence of the so-called "human in the loop" (Howard et al., 2020) is necessary, in particular when the impact of the decision is critical. Humans are capable of deciding when performance of an automatic system is inappropriate or replace machines when the latter fail. This being said, it is interesting to note that there are several broad commonalities between the UVSD and CNNs or other automatic systems. An obvious one is that both rely on a statistical inference, using some kind of a difference variable to compute the similarity between images. Similarity is the critical variable also in PCA and LDA approaches (Kramer et al., 2018). In addition, in both UVSD and automated systems, within-identities and between-identities variability determine the difficulty of the matching task, and those sources of confusion are affected by intraclass and interclass similarity (Rice et al., 2013). For example, to make their benchmark test more challenging for algorithms, Rice et al. (2013) have chosen same-identity images that received low-similarity scores and different-identity images that received high-similarity scores. The current UVSD model can readily account for this manipulation, by assuming that the distribution of match face pairs was moved closer to that of the mismatch distribution and/or by assuming that this cherry-picking procedure altered the similarity relations between images and therefore the underlying variances of the match and mismatch distributions. Another commonality between the UVSD and the CNNs relates to the underlying face spaces that govern the two frameworks. Most CNNs are designed to extract features from images and compute the overall similarity based on these features (Samek et al., 2021). Thus, the face space of these CNNs is ultimately a featural space (Freiwald et al., 2009; Song et al., 2019). In the UVSD approach, the difference between two face images is computed in the form of a similarity signal. I have shown that such a signal can emerge from a feature-matching operation or due to representation of faces in a multidimensional General Recognition space (Ashby & Perrin, 1988). Both UVSD and CNN frameworks apply some kind of decision rule based on the evidence accumulated, and recently, CNNs have been developed that can estimate their level of confidence in their decision (Szegedy et al., 2013).

However, albeit obvious commonalities, there are significant differences between the two. Most of these systems operate as an

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input–output black box with no transparency into their internal mechanisms (but see Samek et al., 2021). Thus, at this stage, their ability to inform cognitive theories of face recognition might be constrained. In addition, these systems require massive training data to ensure generalization from examples, whereas humans see the image of a person only once and can successfully classify it. This point has been succinctly put by Huang et al. (2008): "Human beings are capable of recognizing faces after only seeing one example image, and thus are fundamentally different from algorithms that are only capable of performing matching against a fixed gallery of exemplars" (p. 6). Moreover, within-identities variability (often termed intraclass variation in computer science) still imposes limitations on automatic systems, as noted by Adjabi et al. (2020) in a recent comprehensive review: "Although the existing machine learning/recognition systems have achieved some degree of maturity, their performance is limited to the conditions imposed in real-world applications. For example, identifying facial images obtained in an unconstrained environment (e.g., changes in lighting, posture, or facial expression, in addition to partial occlusion, disguises, or camera movement) still poses several challenges ahead. In other words, the existing technologies are still far removed from the human visual system capabilities" (p. 1). These authors locate the difficulty as residing in the unavoidable presence of within-identities variation: "The intra-class variations are larger than the inter-class variations, contributing to low results in recognition" (Adjabi et al., 2020, p. 39). Moreover, automatic systems can be easily fooled (Nguyen et al., 2015). A seminal study by Szegedy et al. (2013) has shown that altering minute aspects of an image that are imperceptible to humans can lead CNNs to misclassify the correct label of an image, while claiming for high confidence in their wrong decision. This property is inconsistent with human behavior because, as has been shown in the current work, people's accuracy increases as their confidence gets higher.

Dual-Process Theory and SDT of Face Matching

A classic debate in the memory literature pits SDT against alternative dual-process theories based on remember and know sources of information (Donaldson, 1996; Malmberg et al., 2004). A question that arises is whether an alternative account, such as the dual-process signal detection (DPSD; Yonelinas, 1994), can replace the proposed UVSD model of face matching. To apply the DPSD to the face-matching task, we have to consider each member of the face pair as a separate entity and assume that encoding of a face pair on a given trial is held according to a serial architecture (Fitousi, 2021b), in which evidence for either familiarity and/or recognition can be generated. The DPSD assumes that recognition is an all-or-none high-threshold process held with high confidence, while familiarity is a continuous process. Thus, for match trials, a recognition process occurs with probability R, or alternatively, the observer opts for a familiarity-based process that is represented by an equal variance SDT model for which the observer emits a "match" response if the familiarity signal exceeds a criterion c. Thus, the hit rate amounts to $H = R + (1 - R) \times \phi(d'/2 - c_i)$, whereas for mismatch trials, only a familiarity-based process occurs and the probability of a "match" response, and therefore a false alarm, amounts to $F = \phi(-d'/2 - c_i)$. It is well known that the DPSD can fit curvilinear ROC curves, just as the UVSD (Wixted, 2007). Thus, it seems that a dual-process interpretation of the current data is plausible.

Moreover, as noted by Wixted (2007), a dual-process interpretation of the UVSD is also possible:

> In contrast to the DPSD model, the UVSD model holds that each recognition decision is based on a continuously distributed memory strength variable. When memory strength exceeds a criterion value, the item is declared to be old; otherwise it is declared to be new. However, it should not be assumed that the single, unidimensional memory strength variable in the UVSD model necessarily implies a single underlying memory process. Along similar lines, Kelley and Wixted (2001), Wixted and Stretch (2004), and Rotello et al. (2004) argued that the memory strength variable in question consists of the additive combination of familiarity and recollection (or, equivalently, item and associative information). That is, when making a decision about an individual test item, the participant is assumed to take both sources of information into account, much like a juror who combines several sources of evidence into an overall assessment of the defendant's guilt (instead of relying on either one source of evidence or the other). (p. 156)

Can a comparable case be made for face-matching performance? And if so, what arguments can support the hypothesis that multiple psychological systems (sources of information) are not necessary and that a single psychological system (source of information) is sufficient to compute the identity signal that is described in the proposed theoretical framework?

The present study offers, what I believe to be, a first step toward the development of formal models of face matching. Thus, it would be premature to strongly argue against alternative models such as the DPSD. Obviously, more work is needed. Moreover, at this stage, one cannot rule out the UVSD in favor of the dual-process version of the UVSD (Kelley & Wixted, 2001), mainly because the two predict comparable empirical results. This is a mimicry problem that may be solved by future research. However, there are several lines of evidence that do support a single-system continuous signal model of face matching. First, the similarity ratings collected in Experiment 3 exhibited continuous linear relations with the accuracy rates of face matching in Experiment 1b, for both match and mismatch face pairs. This is remarkable, given that these ratings were recorded from two independent groups of participants. This suggests that accuracy rates observed in the face-matching task of one group could be predicted by similarity ratings from another group of participants performing another task that relies on a continuous variable (similarity). Moreover, the similarity ratings in Experiment 3 were linearly related to the confidence ratings in Experiment 1b, which, again, were collected from two independent groups performing different tasks. Unlike the confidence ratings, similarity ratings do not require from the observer a sharp distinction between match and mismatch pairs. Yet, performance in the two tasks was found to be comparable. Thus, similarity ratings are likely reflecting a continuous familiarity-based process that exists in the confidence-rating task. If the face-matching decision with match pairs in Experiment 1b were mostly performed according to a high-threshold process with high-confidence ratings for match face pairs, as the DPSD predicts (Yonelinas, 1994), then no graded and symmetric relations should have been observed between similarity and confidence. In addition, when similarity ratings were treated as if they were confidence ratings (in Experiment 3), and fit to SDT models, they produced comparable results to those obtained in Experiments 1b and 2b with the confidence-rating task. This is commensurate with the idea that the decision variable in these experiments is a continuous similarity signal. Another source of support comes from phenomenology. Although people do not know

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exactly why two faces match or mismatch, they can still perform the task with relatively medium-to-high success rate. This entails that matching is not a high-threshold process, but rather a graded one. In addition, the absence of a mirror effect in face matching with unfamiliar faces (Megreya & Burton, 2007) is consistent with the idea that decision is not governed by two signals (one for each face), but rather according to a difference variable, that once computed, forgets about the individual faces. Moreover, similarity is a fundamental construct in both behavioral and neurophysiological face-space models (Nosofsky, 1988; Valentine, 1991; Vanpaemel & Storms, 2008), and it can explain a great deal of empirical phenomena (Trujillo & Anderson, 2023).

From Unfamiliar to Familiar Faces

The current results mostly pertain to unfamiliar face matching, but the UVSD applies to face matching in general. I believe that the current model can account for matching performance with both familiar and unfamiliar faces. The basic idea is that familiarity is not an all-or-none variable, but rather a continuum (Kramer et al., 2018). Gradual changes in familiarity can be adequately represented by the UVSD as changes in similarity and consequently as changes in the variances of the match distribution and its location relative to the mismatch distribution. It is more parsimonious to assume a single mechanism for both familiar and unfamiliar faces rather than two systems. Moreover, the distinction between familiar and unfamiliar faces is not strict. This is perfectly consistent with the assumption of a continuous similarity signal postulated by the UVSD.

A central question in face research concerns how familiar and unfamiliar faces differ in perception and memory (Hancock et al., 2000). Many researchers postulate that the two are simply unrelated classes of objects, and as such are processed in qualitatively different ways (Megreya & Burton, 2006). However, it is now becoming evident that familiarity is not an all-or-none variable, but rather a graded variable (Burton et al., 2016). A key insight proposed by Kramer et al. (2018) is that variability in the images of the same person (within-identities variability) is essential for learning a facial identity and creating a high-fidelity representation that is resilient to changes in pose, lighting, age, and so forth. Faces become more familiar as the variability of images to which the observer is exposed increases. This proposal is supported by the finding that participants learn a facial identity more effectively when they are exposed to greater variation in images (Ritchie & Burton, 2017; Ritchie et al., 2021).

To study this issue in detail, Kramer et al. (2018) have used computer simulations with PCA and LDA face spaces, testing the relations between familiarity and face matching. First, they have shown that increasing the number of training images (familiarity) increased the system's accuracy of recognition, giving currency to the notion that familiarity is a graded rather than discrete variable. Second, they subjected their computer model to the face-matching task with the faces from the GFMT. Accuracy was computed as the Euclidean distance in the PCA + LDA space between the centroids of each image in the pair, which can be considered also as a similarity measure (Shepard, 1962). The model produced performance levels that were comparable to those of humans. Third, after training the system with certain face pairs, and making them familiar to the system, the distances between two mismatch identities increased, while the distances between images of the same identity decreased. The simulation suggests that familiarity increases the similarity between same-identity images and decreases the similarity between different-identity images. This is an important study because it shows that familiarity is a continuous variable and that levels of familiarity are related to levels of between-identities and within-identities variances.

The simulations by Kramer et al. (2018) reveal that familiar and unfamiliar faces are not two unrelated entities, but rather can be represented on a single face space that is governed by a single similarity metric. Familiarization changes only the perceived similarity relations of the same-identity and different-identity images. The assumption that a single-metric underline faces and that both within-identities and between-identities variances affect performance is also made by the current UVSD framework. Along this vein, the UVSD can therefore be applied to both familiar and unfamiliar faces. The basic idea is that as participants become more familiar with the facial identities in question, the variance of the match distribution decreases because the similarity signal becomes stronger and less variable. It is also possible that the variance of the mismatch distribution decreases too because these face pairs produce weaker and less variable similarity signals. This is exactly what Kramer et al. (2018) have found in their computer simulation study. Such a process can explain why matching familiar faces, in contrast to unfamiliar faces, is considered to be a highly accurate task (Megreya & Burton, 2006). The UVSD can account for the robust observation that matching with familiar faces produces higher rates of accuracy. This can be done by assuming that familiarity reduces the variances of the match and mismatch distributions and/or pushes the distributions apart from each other. One issue that remains to be explained is why familiar faces exhibit the mirror effect, whereas unfamiliar faces do not.

The mirror effect (Glanzer & Adams, 1985) is a general phenomenon in recognition memory. It has been studied with a number of variables, but mostly with word frequency. The main finding is that in an old–new recognition test, low-frequency words produce higher hit rates and lower false alarms than high-frequency words. This phenomenon has been often interpreted in terms of an unequal variance SDT (see DeCarlo, 2007). Researchers assume that low- and high-frequency words evoke separate distributions on the axis of familiarity. The distribution for low-frequency new (LN) words is posited to the left of the distribution for high-frequency new (HN) words, while the distribution for high-frequency old (HO) words is posited to the left of the distribution for low-frequency old (LO) words. In the context of face matching, Megreya and Burton (2007) have argued that a mirror effect exists for familiar faces but absent for unfamiliar faces. Their reasoning was based on the finding of a positive correlation between match and mismatch accuracy at the item level for the familiar faces, but not for unfamiliar faces. Presumably, a positive correlation at the item level is comparable to a mirror effect because an item of a face pair that is readily declared as "match" is also easily judged as a "mismatch." Since such a correlation has not been found with unfamiliar faces, it was taken as evidence that (a) familiar and unfamiliar faces are processed differently and that (b) match and mismatch conditions for unfamiliar faces are governed by two independent systems/processes.

First, it should be noted that the current UVSD assumes a difference signal that reflects the degree of similarity/dissimilarity between members of a face pair. According to this model, once the observer encodes the two faces and a similarity signal has been evoked, the observer forgets about the individual faces he perceived.

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Thus, the model assumes that information about the individual faces is lost and as such is fully compatible with the absence of a mirror effect with unfamiliar faces. Moreover, the absence of a mirror effect is not problematic for SDT frameworks in general. If anything, the emergence of a mirror effect in the context of word frequency came as a surprise for SDT theorists, requiring auxiliary assumptions. But UVSD models have been developed to account for this phenomenon by assuming the existence of separate distributions for high- and low-frequency words (DeCarlo, 2007). Comparable SDT accounts can be developed to explain the emergence of a mirror effect with familiar faces. For example, a variable that is analogous to word frequency and can likely generate a mirror effect in the context of face perception is distinctiveness (Valentine, 2005). Faces that appear as more unique (=low-frequency) have been found to be more efficiently recognized compared to standard faces (=high-frequency). A second possibility is that familiarity breeds attention to individual faces (Yonelinas et al., 1999), and as a result, participants encode not only information about the relations between faces but also about the actual faces. This allows for the emergence of a mirror effect. A third possibility is that canonical face-matching tests with unfamiliar faces are constructed in such a way that eliminates the perceptual salience of distinctive faces by pairing distinctive faces in the mismatch condition only with other distinctive faces. This procedure is completely justified experimentally, perhaps in an attempt to enhance the difficulty of matching. But if faces were paired randomly, then the salience of distinctive faces could have been expressed more prominently. According to the unified confidence–similarity UVSD model, this would imply that face pairs that consist of a distinctive face would generate a weaker similarity signal across match and mismatch conditions, leading to a mirror effect with unfamiliar faces as well.

Conclusions

The ability to match faces correctly is crucial for efficient face recognition. It also plays an important role in applied setting such as in police investigations, passport control, and eyewitness memory. The present study outlines a theoretical framework for interpreting central aspects of face matching. The model is dubbed the unified confidence–similarity UVSD model. As its name indicates, this model can explain a myriad of phenomena documented in the literature, along with a set of new predictions concerning: (a) accuracy, (b) confidence, (c) the relations between confidence and accuracy, (d) perceived similarity, (e) the relations between similarity and accuracy, and (f) the relations between similarity and confidence. The model was tested in three experiments and was supported by substantial evidence.

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